Agricultural credit and women's agency: Experimental evidence from India

Tarana Chauhan*[1], Berber Kramer[2], Patrick S. Ward[3], and Subhransu Pattnaik[4]

[1]Brown University, [2]International Food Policy Research Institute, [3]University of Florida, [4]University of Göttingen

June 26, 2026. Link to latest version.

Abstract

Women-focused programs raising income and agency are costly and difficult to implement, and evaluations show that targeted cash transfers and loans are often appropriated into assets and enterprises outside women's control. We hypothesize that improving liquidity constraints for households increases household consumption and residual income for individual members, and evaluate its consequences for women. We test this using randomized provision of collateral-free loans for farming across 80 villages in India. Employing ANCOVA and IV estimations, we show that the loan expanded household resources through investment in productive capital and consumer durables; women's financial inclusion improved substantially, driven by smartphone ownership and mobile banking use; women's time allocation shifted away from farm labor toward personal activities, consistent with productive capital relaxing demand for female labor. The results demonstrate that a non-targeted household-level program can improve women's outcomes through resource expansion without affecting their involvement in intrahousehold decision-making.

Keywords: Digital finance, Smallholder farming, Women's Financial inclusion, Labor supply; **JEL codes:** J12, O55, O57, Q54

---

*Corresponding author: Tarana Chauhan, Postdoctoral Research Associate, Center for Philosophy, Politics, and Economics, Brown University, email: tarana_chauhan@brown.edu. See CRediT authorship contribution statement at the end. This paper evaluates data from an experiment registered with the American Economic Association's registry for randomized controlled trials (Kramer and Ward, 2023; https://doi.org/10.1257/rct.11616-1.0). We are grateful to the implementing partner Dvara E-Registry for facilitating an impact evaluation of their product. This project was funded by the Center for Effective Global Action (CEGA)'s Digital Credit Observatory, a Gates Foundation-supported initiative.

1 Introduction

Programs targeting resources to women don't always succeed in improving women's economic outcomes or agency within the household (Banerjee et al., 2015; Green et al., 2015; Attanasio et al., 2015; Angelucci et al., 2015; Said et al., 2019). Evaluations of loan programs to female entrepreneurs and cash transfers find evidence of resource appropriation (Fiala, 2018; Bernhardt et al., 2019) and male backlash (Angelucci, 2008; Bobonis et al., 2013) in several contexts, highlighting the potential risks to women through these targeted interventions. Programs that succeed in increasing women's decision making ability and earnings incorporate mechanisms that give women direct control over these resources (Fiala, 2017; Riley, 2024). However, such programs are costly to implement and difficult to scale up.

In this paper, we show that a program that was not designed for women improved their outcomes through resource expansion by reducing household's liquidity constraints. We evaluate collateral-free loans to 1,429 farming households in 80 villages of India using data from a randomized controlled trial. The loan product was offered between 2023 and 2024 in 40 of the study villages (treatment group), while the remaining 40 villages (control group) did not receive this product but could access formal and informal credit suppliers in the market. The program loan differed from other available formal loans with its dual features of low interest rate and no requirement of collateral. In addition to investigating the impact of the loan product on household investment and consumption, we evaluate how it changed women's financial inclusion, time use, and involvement in intrahousehold decision-making.

We anticipate changes in women's outcomes as agricultural production is organized at the household level in India with gendered task specialization[1]. Targeting loan to a particular household member in this context of joint production may not impact how households use these funds. However, expanding the household budget in the short term can benefit household members in two ways. First, if the loan increases household's consumption of (composite) goods that are shared or privately consumed, such as productive assets and consumer durables, it will increase members' utility. Second, individual members may adjust their private consumption and labor allocation in response to loan expenditure on composite goods and any changes in their residual income.

We employ two empirical models in our analysis. First, we estimate the intent-to-treat (ITT) effect of the loan program on household welfare and agricultural intensification

[1] Most farm managers are men, and women typically undertake labor-intensive activities such as weeding, harvesting, and livestock management (Afridi et al., 2023; Gulati et al., 2025).

using an analysis of covariance (ANCOVA) model. Here we exploit the cluster randomization of the loan program at the village level and a household panel data combining baseline and endline surveys. Second, we examine how the loan affected the well-being of household members, focusing on their financial inclusion, time use, and involvement in decisions. The endline survey interviewed two members per household, compared to one at baseline - the farmer that was interviewed in the baseline survey and one additional household member, yielding a sample of more than 1,000 women. We use an Instrumental Variables (IV) model in which randomized village-level treatment assignment instruments for households receiving the loan product. Restricting the endline sample to female respondents and controlling for their position in the household, we estimate the Local Average Treatment Effect (LATE) of the program on women's outcomes. We apply the same empirical model to men in our sample.

The analyses yield four key findings. First, farming households use the loan money to accumulate assets and consumer durables. They increased ownership of non-mechanized farm equipment by 14 percentage points (pp) and of large consumer durables by 8 pp for households in the treatment arm relative to the control arm. They also invested in business equipment but did not change spending on other agricultural inputs such as seeds, fertilizers or labor[2].There were no impacts on real income from farming but the program led to a significantly greater food consumption score than the control group (39.4 vs. 33.8). Similar to a cash transfer program, farmers used savings from a cheaper loan on consumer durables and food security (Haushofer and Shapiro, 2016).

Second, the loan positively impacted household members' financial autonomy. Women were 81 pp more likely in loan product-receiving households to use the debit cards linked with their individual bank account solely than the control group. They were also 94 pp more likely to use mobile apps for banking. The significantly greater uptake of digital financial services in the treatment arm is driven by smartphone ownership (24 pp). These results suggest that women individually captured part of the resource expansion. The loan had a positive impact on men's financial autonomy as well but the effect sizes are smaller given higher control group averages.

Third, there are shifts in women's time allocation that correspond with large aggregate increases in households' ownership of productive capital and consumer durables. Women in loan product-receiving households allocate significantly more time to personal activi-

[2] Kramer et al. (2025) evaluate the effects of the credit product on agricultural investment for all study villages in the intervention. This paper re-estimates some of the same variables to verify theoretical predictions for how household decisions impact individual welfare. Any differences in aggregate effects and interpretation of agriculture-related outcomes are from differences in sample and estimation methods.

ties, such as leisure (2.6 versus 1.7 hours in the control group). Women's participation in farming is significantly less than control, while men's labor supply remains similar. In the absence of aggregate changes in harvest or hiring labor, this pattern indicates that the acquisition of productive capital may have specifically relaxed the demand for female labor. There were no changes in time allocation to domestic work and time spent in non-farm economic activities declined, driven largely by significant reduction of any non-farm employment. Men's time allocation to production did not differ significantly but they allocated more time to personal activities.

Finally, we do not find an impact of the loan on women or men's intrahousehold bargaining power through decision making. To the extent that involvement in these decision domains reflects intrahousehold bargaining power, the results suggest the resource expansion did not shift the balance of power between spouses. The loan reinforced existing cooperative household arrangements rather than creating new ones. There are heterogeneous effects by women's ex-ante bargaining power: Greater spousal coordination among households prior to the intervention corresponds positively with women's involvement in decisions on non-agricultural production activities, domestic work, household purchases, their private consumption and personal activities.

We examine heterogeneous treatment effects on individual welfare by household's wealth and socio-economic identity and find no differential trends in women's financial autonomy or time allocation to personal activity. These results are also consistent Odisha and Maharashtra, though effects on women's time use in farm and non-farm work and involvement in household decisions vary by state, as we discuss in Section 7.

This paper contributes to three strands of literature. The first examines how women's access to financial resources and digital tools shapes their control over those resources. Women's uptake of digital finance remains systematically low in rural India, partly because gender norms restrict unmarried women's access to the internet (Barboni et al., 2018). Interventions aimed at expanding women's access to digital tools have struggled to overcome these constraints. In a government program that distributed smartphones to rural women, 40 percent transferred the device to another household member, and women's knowledge and use of mobile banking did not improve (Barboni et al., 2024). In contrast to this earlier evidence, an effect of the loan program was greater smartphone ownership by women as well as knowledge and use of mobile banking.

A broader literature similarly finds that expanding women's financial access often does not translate into greater agency within the household. Access to microfinance, savings groups, business loans and grants have had limited impact on women's decision making ability (Banerjee et al., 2015; Karlan et al., 2017; Angelucci et al., 2015; Field et al., 2021). Loans to women in multi-adult households have raised overall household income and profits of other members' enterprises but not women's own profits (de Mel et al., 2008; Garikipati, 2008; Fiala, 2018; Friedson-Ridenour and Pierotti, 2019; Bernhardt et al., 2019). Evidence from Uganda shows more promising effects where women invested in their own business when they received the loan money in their mobile accounts (Riley, 2024). In our paper, we find that increased liquidity from the loan program improved women's digital finance use, but their involvement in household decision making did not rise, consistent with the broader literature.

Second, this paper adds to the literature on labor supply response to transfers or agricultural credit. Fink et al. (2020) reported increased labor supply in response to loan while transfers reduced women's probability of working and increased men's participation in domestic chores (Hidrobo et al., 2016; Beaman et al., 2023; El-Enbaby et al., 2025). In our paper, we show that both women and men increase their time allocation to leisure. Women reduce their time allocation to farming and men spend more time on domestic chores.

Third, this paper contributes to the complementary literature evaluating the effect of cash grants and microcredit on agricultural investment (Karlan et al., 2014; Tarozzi et al., 2015; Maitra et al., 2017; Nakano and Magezi, 2020; Fink et al., 2020; Beaman et al., 2023; Mukherjee et al., 2024). While there are no direct improvements in farm productivity or incomes in our sample (see Kramer et al. 2025 for an evaluation of the program on agricultural investment), households increased ownership of farm and business equipment and consumer durables. Additionally, this paper yields important policy implications as it shows unintended, positive consequences of the local product on women's outcomes. This is relevant as development finance institutions and governments are actively trying to scale up such products in low-and-middle income countries. Household-level interventions can therefore be utilized as cost-effective ways to improve women's outcomes while avoiding the pecuniary and social costs of targeted programs.

2 Theoretical framework: Loan and Individual Welfare

Agriculture in India involves joint contribution by household members on the same plot. The Indian Ministry of Agriculture & Farmers Welfare formally recognizes this, classifying agricultural land managed by members of the same household as "individually operated". Doss (2018) documents that it is difficult to separate individual contribution to and profits from production. Targeting agricultural credit to women in farming households with multiple adults, therefore, may not guarantee their autonomy or participation in decisions around the use of loan money. There is limited evidence to verify whether the returns from focused agricultural loans to women in multi-adult households are higher than loans to the household.

Given collaboration between household members in decisions around farm management, one way to predict household resource sharing would be through a unitary model where the household has a single decision maker and household members have homogeneous preferences. Under the assumptions of this model, household's use of loan money would not depend on which member is the beneficiary of the product (Samuelson, 1956). The addition of loan money would increase non-labor income of the household and consequently their consumption. We explore a more flexible framework instead where household members have different preferences (Chiappori, 1988). By extending the collective household model in Bobonis (2009), we allow the loan to affect individual welfare through both household-level and private consumption channels. It also helps distinguish the differential impact of the loan on different household members.

2.1 Set-up

Consider a two-member household with members $i \in \{A, B\}$. Each member derives utility from private consumption goods $q^i \geq 0$, leisure $l_i \geq 0$, and a private asset position $\alpha_i \geq 0$ that represents financial autonomy. The household jointly consumes a public good $K \geq 0$ and a composite good $C = C^A + C^B + C^H \geq 0$, where C^A and C^B denote the shares consumed privately by each member and C^H denotes the jointly consumed share. Consumer durables and productive capital are examples of composite goods that generate utility for both members.

Each member is endowed with time T and faces a wage rate w^i, yielding labor income $w^i T$. Individual non-labor income is y^i, and the household receives joint non-labor income y^O. Total household resources are:

$$Y = \sum_{i \in \{A,B\}} \left(y^i + w^i (T - l_i) \right) + y^O \tag{1}$$

Member i's preferences are represented by a utility function $U^i(q^i, l_i, \alpha_i, C, K)$, which is strictly increasing, twice continuously differentiable, and strictly quasi-concave in all arguments. We impose three assumptions that are relevant to the program context.

Assumption 1. *The household treats the loan $\mathcal{L}$ as an increase in joint income in the short term:* $y'^O \mapsto y^O + \mathcal{L}$.

Assumption 2. *The loan has a strictly monotone influence on composite good consumption:* $\partial C/\partial y^O > 0$*.*

Assumption 3. *Each member has a strict preference for leisure and financial autonomy:* $\partial U^i/\partial l_i > 0$ *and* $\partial U^i/\partial \alpha_i > 0$*.*

Assumption 1 reflects the context that farm production is organized at the household level and the program design where two household members co-signed the loan. Therefore, a loan for farming is likely to be treated as a joint resource. Assumption 2 ensures that increases in joint resources translate into observable gains in composite good consumption, providing the household-level channel through which the loan benefits all members. Assumption 3 states that both leisure and financial autonomy are strictly valued.

2.2 Household Optimization

Under Pareto optimality, the household's allocation is characterized as the solution to a weighted utility maximization. Let $\mu \in (0, 1)$ denote member A's Pareto weight, making $1 - \mu$ as member B's weight. The household solves:

$$\max_{\{q^i, l_i, \alpha_i, C, K\}} \quad \mu\, U^A(q^A, l_A, \alpha_A, C, K) + (1-\mu)\, U^B(q^B, l_B, \alpha_B, C, K) \tag{2}$$

subject to the household budget constraint:

$$\sum_i \left(p_q q^i + p_\alpha \alpha_i + w^i l_i\right) + p_C C + p_K K \le Y \tag{3}$$

and the non-negativity constraints $q^i, l_i, \alpha_i, C, K \ge 0$ for all i.

The Pareto weight μ is not fixed but depends on the distribution of resources within the household. Following Browning and Chiappori (1998) and Bobonis (2009), we allow μ to be a function of the distribution factors. These are variables that affect bargaining power without entering preferences directly: $\mu = \mu(y^A, y^B, y^O)$ with $\partial\mu/\partial y^i \neq 0$ and $\partial \mu/\partial y^O \neq 0$. Under this formulation, the loan affects intra-household allocations through two distinct channels: it relaxes the household budget constraint (an income effect common to both members) and it shifts the Pareto weight (a distributional effect that reallocates resources between members).

2.3 Two-Stage Budgeting

The optimization problem includes a two-stage budgeting that distinguishes the loan's distinct effects on household-level and individual-level consumption.

Stage 1. Given total household resources Y and the Pareto weight μ, the household chooses expenditure on the public and composite goods. Let $C^*(Y,\mu)$ and $K^*(Y,\mu)$ denote the Stage 1 optima. By Assumption 2, $\partial C^*/\partial Y > 0$, the loan raises composite good consumption for the household as a whole. Because both members value C and K, this channel generates a welfare gain for all members irrespective of how the loan affects bargaining.

Stage 2. Conditional on C^* and K^*, each member i chooses private consumption to maximize utility subject to an individual residual income constraint. Member i's residual income is:

$$\rho^i = y^i + w^i T - s^i \tag{4}$$

where $s^i \geq 0$ is member i's contribution to Stage 1 household expenditure, determined by an intra-household sharing rule $s^i = \phi^i(\mu, Y)$ with $\sum_i s^i = p_C C^* + p_K K^* - y^O$. Member i solves:

$$\max_{\{q^i, l_i, \alpha_i\}} \quad U^i(q^i, l_i, \alpha_i, C^*, K^*) \quad \text{s.t.} \quad p_q q^i + p_\alpha \alpha_i + w^i l_i \leq \rho^i \tag{5}$$

with C^* and K^* taken as given. The solution to (5) yields member i's indirect utility $V^i(\rho^i, C^*, K^*)$, which is strictly increasing in residual income, composite and public goods. The total effect of the loan on member i's welfare decomposes as:

$$\frac{dV^i}{d\mathcal{L}} = \underbrace{\frac{\partial V^i}{\partial C^*} \cdot \frac{\partial C^*}{\partial \mathcal{L}}}_{>0} + \underbrace{\frac{\partial V^i}{\partial K^*} \cdot \frac{\partial K^*}{\partial \mathcal{L}}}_{>0} + \underbrace{\frac{\partial V^i}{\partial \rho^i} \cdot \frac{d\rho^i}{d\mathcal{L}}}_{\gtreqless 0} \tag{6}$$

The first two terms are unambiguously positive by Assumption 2 and the strict monotonicity of V^i. The sign of the third term depends on how the loan affects member i's residual income, which in turn depends on the sharing rule response to the change in y^O.

2.4 The Effect of the Loan on Residual Income

The loan affects individual residual income in two ways. One is a bargaining channel, where if the loan shifts the bargaining rule in member i's favor, their required contribution s^i falls and ρ^i rises. The other is the resource expansion channel. As total household resources y^O grow, the total individual contributions required to finance joint expenditure fall, relaxing both members' private budget constraints. We characterize the welfare implications of the loan on individuals by considering three cases.

Case (i): $d\rho^i/d\mathcal{L} > 0$. The loan increases member i's residual income, either because it strengthens their bargaining position or because the income expansion channel reduces the burden of household expenditure contributions. All three terms in (6) are strictly positive. The loan is unambiguously welfare-improving for member i. By Assumption 3, the expanded residual income allows member i to increase leisure and financial autonomy above their pre-loan optima, generating welfare gains on all private consumption margins.

Case (ii): $d\rho^i/d\mathcal{L} = 0$**.** The loan leaves member i's residual income unchanged, as the bargaining and income expansion effects on s^i offset one another or neither operates. The welfare effect in (6) is positive as the first two terms are positive and third term equals 0. Member i benefits from the household-level consumption gains but experiences no change in private consumption, leisure, or financial autonomy. This case is consistent with a household that allocates the loan entirely to joint expenditure without altering the internal distribution of private resources.

Case (iii): $d\rho^i/d\mathcal{L} < 0$**.** The loan reduces member i's residual income, as the shift in bargaining power or the income expansion channel increases their required contribution s^i. The net welfare effect is now ambiguous. When the residual income loss is small relative to the gains in jointly consumed goods, $\Delta V^i > 0$ and the loan remains welfare-improving. When the loss is large enough to dominate, $\Delta V^i < 0$. Case (iii) where a positive household income shock leaves an individual member worse off cannot be detected in a unitary model, further motivating the collective framework.

2.5 Testable predictions

We analyze the first stage changes in household consumption through measures of asset ownership and food security. For the second stage, we do not observe individual residual income ρ^i, however, we are able to infer welfare effects of the loan on household members through changes in labor supply and financial autonomy. We infer any changes in the bargaining weight μ through individual reports of involvement in decisions around household and personal activities.

3 Context, Program Description and Evaluation

3.1 Study population: Credit-constrained farming households

This intervention was conducted in the districts of two Indian states — Jajpur in Odisha and Amravati in Maharashtra (Figure 1). The main crops cultivated in Jajpur are paddy (rice) and in Amravati, cotton and soybean. These districts were identified by our collaborator Dvara e-registry (DER) as either regions where farmers were credit constrained, DER had worked with farmers on other initiatives previously, or both. The study targeted loan provision to small and marginal farmers[3] which constitute the majority of agricultural holdings in both states. As per the Agricultural Census 2015-16, over 30%

[3] Farms of size between 1 and 2 ha are classified as "small", and less than 1 ha as "marginal".

of Maharashtra's and Odisha's landholdings are classified small. The share of marginal landholdings in Odisha and Maharashtra is 45% and 18% respectively.

Irrigation is predominantly rainfed in both Jajpur and Amravati. Paddy is typically transplanted June through July and harvested in September through November. The agriculture sector in Odisha employs more than 70% of rural women, most of whom are engaged in rice farming (Barooah et al., 2025). Women almost exclusively perform nursery preparation, transplanting, weeding, harvesting, and threshing. Weeding is done manually or with simple hand tools. Cotton is sown from May to June in Maharashtra and harvested between October through December. For cotton, women cultivators in Maharashtra are responsible for the majority of on-field tasks such as sowing seeds, stalk removal, ensuring optimum plant population, weeding, fertilizer application, and picking (Dwivedi and Damle, 2019). Men are primarily responsible for operating machinery (tractors, ox-carts) and for managing market and sales transactions. Soybean is sown June- July and harvested September through October. For soybean, women perform the more labor-intensive manual tasks including weeding, loosening soil, and seed preservation, while men handle ploughing, marketing, and machine operations.

The key constraints for small and marginal farmers in our study locations are expensive interest rates and the absence of land title. The main sources for formal credit are - commercial banks (government or privately owned), small finance banks (private sector entities servicing small and marginal producers), microfinance institutions (MFIs), cooperatives, self-help groups (SHG) and registered entity of producers that share profits called farmer producer organizations (FPO). Interest rates of loans offered by MFIs and small finance banks vary from 18 to 26% in Odisha and is about 24% in Maharashtra. The interest rate on loans from commercial banks is significantly cheaper, but these loans are prohibitive because of collateral and document requirements and delayed loan processing. Loans from SHG and FPO do not require collateral for members but are available on rotation and not every planting season. The Indian government offers a collateral-free loan of upto INR 200,000 (about USD 2,250) to farmers called the Kisan Credit Card (KCC) at the maximum interest rate of 7%, however, it is restricted to farmers that own land. Under KCC, short-term credit is provided by commercial/ small finance banks or cooperatives, and the loan amount is based on pre-determined regional terms, size of land, harvest, depreciation and insurance requirement. Figure A.1 shows that fewer than 20% of farmers in our sample ever took a KCC loan, consistent with the fact that less than half owned land (47%), a prerequisite for KCC eligibility.

3.2 Loan program and Eligibility

The intervention offered collateral-free loan to small and marginal farmers at 14% interest. The program implementer, DER, partnered with one commercial and one small finance bank to provide these loans in the summer planting season in 2023 and the winter season spanning 2023-24[4]. In this paper, we only analyze agricultural investment in the summer planting season as most farmers in the surveyed regions do not cultivate during winter. The loan amount varied by size of land and whether the farmer was first time/ repeat borrower during the intervention. It ranged from INR 30,000 to 60,000 (USD 330 - 660). The loan was offered at the beginning of planting season and required repayment in monthly installments[5]. The program addressed two additional challenges to credit access for these farmers - loan application and processing times. DER agents visited farmers at their field or home and filled out the application on their behalf. The loan was processed within 7 business days for successful applicants.

To be eligible for the loan, farmers were required to be residents of their village for at least five years, operating on land no more than 3 acres, and had cultivated at least once in the past three years in the agricultural season they were borrowing for. They were also required to have attended a meeting organized in their village at the start of the intervention (see 3.3). Their credit worthiness was computed by DER using key parameters from the last 3 years that were expected to affect the farmer's cash flow. These are crop productivity, farm area, damage from natural calamities or other factors, soil moisture and irrigation availability, soil nutrients and vegetative growth. Several of these variables were measured using remote sensing tools. Loan applicants were expected to have a credit score above 40 in this weighted index. Figure A.2 shows that almost all the farmers in the study were above this threshold. Evaluations from established credit bureaus were also assessed when available.

Cost of program

The program included fixed costs such as project management costs, field office and other materials that are listed in Table A.1. In addition, there were some variable costs such as calculation of the credit score for each farmer, engagement of field staff and discretionary allowances. The loan had 14% interest rate which resulted in earnings of USD 63-72 per farmer. Subtracting total costs per farmer from the total revenue through interest payments yielded a profit of about USD 13 and 23 per farmer in Odisha and Maharashtra, respectively.

[4] Due to operational constraints, the program was not effectively implemented in winter season 2022-23.

[5] The implementing partner reported that farmers expressed preference for a monthly repayment plan over lumpsum payment after harvest.

3.3 Randomization

The experimental sample comprised 80 villages identified by DER: half were located in two blocks of Jajpur district, Odisha, and the remaining half in four blocks of Amravati district, Maharashtra.[6] DER previously operated in some of these villages through supply chain management and livestock provision. However, they did not supply loans in the selected villages before the program. After consultation with village heads, DER organized meetings in each village on a day when farmers were not expected to be in their fields (September through October in 2022). DER discussed credit options available to the farmers in these meetings and did not elicit applications or describe terms of their product in these meetings. It also did not include any information that could change social norms around loans, use of mobile phones, financial autonomy and individual welfare. The villages were randomly assigned into treatment and control arms by lottery *after* the conclusion of meetings in all 80 villages. Half the villages (20 per state, 40 in total) were assigned to the treatment arm and the remaining half to control (split equally between the 2 states). A roster of at least 20 farmers was created for each village by eliciting names and contact information from participants of the information session.

4 Data

4.1 Data collection and Attrition

Baseline: The survey was conducted between November 2022 and February 2023 by an independent team of enumerators hired and trained by co-authors of this paper (Kramer, Ward and Pattnaik). The survey interviewed the farmer who attended the information session. The questionnaire included detailed information about the crops cultivated during the summer and winter agricultural cycles, inputs utilized in cultivation, asset ownership, economic shocks, food security and participation in decision making. Across the 80 villages, the goal was to interview approximately 21 households per village and the baseline data included 1,682 households.

Administrative data: We utilize DER records on the number of households that applied for a loan in the treatment arm, were successful and the loan amount they received during the summer of 2023. We also analyze the credit score estimated by DER in 2023 for each household our sample.

Endline: The endline survey, conducted again by an independent team of enumerators between April and August 2024, successfully re-interviewed 1,431 households from the baseline. The survey was expanded to interview two members of each household: the

[6] Blocks are administrative sub-divisions of districts used for planning and development in rural areas. The blocks in Odisha are Dashrathpur and Jajpur; those in Maharashtra are Amravati, Ashti, Morshi, and Tiosa.

farmer who had attended the information session before baseline (henceforth, the "panel respondent") and one additional household member. This second member was successfully interviewed in 1,066 households. We expanded the modules from the baseline survey to also measure respondents' access to financial services and their time use.

Sample size and Attrition: There is 15% attrition between the baseline and endline surveys and the sample is balanced across the intervention arms (column 1, Appendix Table A.2). Column 2 of the same table shows that households' ex-ante characteristics are generally uncorrelated with attrition, with two exceptions: households with a female panel respondent were more likely to drop out of the sample, and those with an outstanding loan were less likely to do so. We omit households where the respondents were below 18 years of age. This gives us a balanced panel of 1,429 households that includes 1,061 other household members.

4.2 Balance test and Summary Statistics

We report household and panel respondent characteristics from the baseline survey in Table 1. Column 2 and 3 also report averages for treatment and control groups, respectively. Households in both intervention arms were statistically similar on observables before the loan program was implemented. The average respondent interviewed in baseline was middle-aged (about 43 years old) and over one-third were women. Less than half (47%) of the households owned land. Borrowing is prevalent - over 40% of the sample took a loan in the 12 months prior to baseline survey, and one in five households had an outstanding loan. Over 90% of panel respondents had a bank account.

4.3 Outcome variables

Household outcomes

Our primary household-level outcomes are asset accumulation and food security that measure changes in household welfare. We also report household investment in agriculture to understand how households used the loan money[7] and impacts on individual outcomes. These include variables such as area planted, real income, yield, harvest, expenditure on inputs (seeds, fertilizers, pesticides, renting machinery and labor), and labor utilization.

Individual outcomes

The individual level outcomes are categorized under financial inclusion, time use and involvement in decision making.

[7] See Kramer et al. (2025) for a formal evaluation of the effects of the program on agricultural intensification.

Financial inclusion: We expect if loan money was apportioned to different household members for consumption, it would affect their bank account activity and use of tools for transactions such as ATM cards and mobile banking. We asked respondents whether they owned an individual bank account, used the debit card linked with this account on their own or shared it with their spouse/ other family members, the frequency of transactions from this account, and knowledge and use of mobile apps for banking. These binary indicators take the value 1 when individual responded yes, and 0 otherwise. Schaner (2016) found that debit cards are perceived to lower transaction costs of resource appropriation from individual bank accounts, and women reduced their account usage when provided a debit card. We investigate whether members are sharing their debit card or using it autonomously to infer their financial autonomy and control over resources. In addition, we measure knowledge and use of mobile app-based banking that lowers both costs of privacy and transactions by removing the requirement to physically travel to a bank branch or ATM for deposit or withdrawal. These are growing in popularity through low-cost internet plans and mobile phones[8].

Time use: Hidrobo et al. (2016); Fink et al. (2020); El-Enbaby et al. (2025) show that loans and transfers to farming households impact time use of members. We expect that any changes in individual time use in response to more resources reflects preferences for that activity. We directly measured how household members spent their time in the past day and during a busy day in the agricultural season ("peak time") on various production activities such as farming, horticulture, livestock management, poultry, fishpond, and non-farm economic activities such as business, self-employment and wage employment. We also investigated non-production activities such as household purchases, domestic work and personal activities. The activities included under each outcome variable is described in Table 2.

Decision-making: We follow the literature to estimate effects on intrahousehold bargaining through self-reported survey measures of decision making (Allendorf, 2007; Connelly et al., 2010; Swaminathan et al., 2012; Field et al., 2021; Annan et al., 2021; Kosec et al., 2022; Heckert et al., 2023; Doss et al., 2022; Quisumbing et al., 2023a,b). We asked questions about members' involvement in the last 12 months in decisions around production and non-production activities such as farming, livestock and poultry, non-agricultural work and consumption (see Table 2 for description). These questions were adopted from the project-level Women's Empowerment in Agriculture Index (pro-WEAI). We measure "involvement" as a combination of solo and joint decision making with other household members as we did not collect individual-level information of preferences on different decisions[9]. We also test separately women's decision making jointly with their

[8] Digital transactions have been steadily promoted by the government of India and Reserve Bank of India (CGAP, 2019)

[9] Kochar et al. (2022) highlights that some decisions, such as those concerning food preparation, align

spouse.

Loan provision was not randomized by gender, therefore, we do not test women's uptake of credit in response to the program.

5 Estimation

5.1 Intent-to-Treat

We estimate the intent-to-treat (ITT) effect of the loan intervention on household resource allocation using the ANCOVA model below. This exploits the random assignment of the intervention and a balanced panel of household outcomes and covariates measured using the baseline and endline surveys.

$$Y_{hvb,1} = \alpha_0 + \alpha_1 Y_{hvb,0} + \alpha_2 TREAT_v + \sum_{j=1}^{J} \theta_j \mathbf{X}_{jhvb,0} + \eta_b + \tau_m + \epsilon_{hvb} \tag{7}$$

All outcomes are measured at the household level h in this model. Equation 7 estimates the effect of the program on the outcome variables while including baseline measures of these variables, $Y_{hvb,0}$, as independent variables. We verify that outcome variables at baseline are balanced between the treatment and control groups (Table A.4), except for the total area cultivated under main crop and total expenditure on male and female labor in the summer season. We control for household covariates described in Table 1 as well as features of soil quality such as moisture, nutrient and productivity that were used to determine each plot's credit worthiness in $\mathbf{X}_{hjvb,0}$[10]. The binary indicator $TREAT_v$ identifies villages v in the treatment $(= 1)$ and control arms $(= 0)$ and coefficient α_2 estimates the ITT effect of the intervention on outcome. Multiple villages constitute an administrative block b and the specification includes block fixed effects (η_b) that control for differences in markets and institutions. Any differences in recall due to survey month of the baseline and endline are controlled for using month fixed effects (τ_m). There are 80 villages in the sample and standard errors are clustered at the level of treatment assignment which is village.

We also use this model to validate the use of the loan for agricultural intensification. While Kramer et al. (2025) investigate the impact of the loan program on agricultural investment and productivity (and disaggregate impacts by ex-ante credit constraint and

with gender stereotypes about women's role in the household, and that delegating these tasks to women may reflect their lower agency

[10] We report estimates conditional on all covariates as there are very small differences in the magnitude of treatment effects when only a subset of covariates identified by post-double selection LASSO method (Belloni et al., 2013) are included.

state), we repeat this exercise in Appendix Tables A.5 - A.8 to document any changes in household production that may affect individual welfare, which is the primary focus of this paper. Our analysis differs from Kramer et al. (2025) in three ways. First, the sample in this paper is restricted to villages in the loan treatment and control arms while Kramer et al. (2025) evaluate a third intervention arm as well. Additionally, we analyze a balanced panel of households in the loan treatment and control villages as attrition is balanced across arms. Second, we explore the effects of the loan program on agricultural intensification only in the 2023 summer planting season although the program also covered the winter season of 2023-24. This is done as most farmers in our study did not cultivate during the winter season. Third, we separately analyze agricultural outcomes for the primary crops grown in our study regions (rice in Odisha and cotton and/or soybean in Maharashtra).

5.2 Local Average Treatment Effect

We analyze the local average treatment effect of the loan on individual-level outcomes using an Instrumental Variables (IV) model. The first stage tests the predictive power of the exogenous assignment of loan treatment, $TREAT_v$, to household h receiving the loan product for cultivation in summer of 2023.

$$ReceivedLoanProduct_{hv} = \beta_0 + \beta_1 TREAT_v + \sum_{j=1}^{J} \theta'_j \mathbf{Z}_{jihvb,0} + \eta_b + \mu_{hv} \tag{8}$$

Table 5 reports that households in treated villages were 48.1 percentage points more likely to receive the loan product. The exogenous treatment assignment corresponds with 48 pp more likelihood of receiving the loan product. The F statistic is 18.6, above the conventional threshold of 10, and the instrumental variable explains approximately 30% of the residual variation in loan take-up after partialling out baseline covariates and block fixed effects. The equation below is the Two-Stage-Least-Squares (2SLS) estimation of the instrumented variation of household receiving the loan product on outcome variables.

$$Y_{ihvb} = \gamma_0 + \gamma Received\widehat{Loan}Product_{hvb} + \sum_{j=1}^{J} \theta''_j \mathbf{Z}_{jihvb,0} + \phi RespType_{ihvb} + \eta_b + \upsilon_{ihvb} \tag{9}$$

We analyze outcomes for female and male respondents separately using this 2SLS model. We control for household characteristics described in Table 1 and individual characteristics such as age and whether respondent is married in $\mathbf{Z}_{hjvb,0}$. We also control for whether the respondent is the panel respondent or the other household member ($RespType_{ihvb}$) in equation 9. We include block fixed effects (η_b) and cluster standard errors by village.

The model satisfies the following key assumptions of an IV model. First, we satisfy that the treatment assignment is exogenous by verifying that household's observable characteristics are balanced at baseline between the treatment and control arms (Table 1). Attrition of households is also balanced across arms (Table A.2). Second, the exogenous village level treatment assignment positively affects household's take up of the loan product (Table 5). Third, the village meetings did not share information that could impact preferences for loans, social norms on financial autonomy and leisure or involvement in decision making. Fourth, we verify implementation fidelity in Table A.3 showing that only households in the treatment arm received the loan product. Finally, any household that met the eligibility criteria listed under section 3.2 could apply for and receive the loan. By this rule, one household's receipt of the loan product did not affect the likelihood of another household in the treatment arm receiving the product.

6 Household responses to the loan program

In this section, we report the ITT estimates of changes to households' spending and income in response to the program. The intervention increased the ownership of productive capital and consumer durables by households in the treatment group (Table 3). There was a 13.8 pp increase in the ownership of farm equipment such as hand tools, animal-drawn plows and non-mechanized irrigation tools in the treatment arm relative to control. Households' investment in equipment for non-farm business activities such as sewing machines increased by 7.1 pp. Households in the treatment arm were 7.8 pp more likely to own large consumer durables. This includes time saving kitchen appliances such as refrigerator, TV, and washing machine as well as other household durables such as sofa, dining table, chair, mattress, cot or bed and computer. We don't find increases in mechanized farm inputs such as tractors, electric pumps, seeder and threshers. Even with subsidies from the government, some of these mechanized equipment cost at least 4 times more than the loan amount. Therefore, the changes in productive capital and consumer durables align with the increase in spending capacity of the farmers in response to the loan.

We also find relative improvements (about 17%) in the food security of the households in the treatment group in Table 4. Panel A measures household food security using a 7 days recall measure of 8 food groups designed by the World Food Programme[11]. We analyzed whether the individual respondent consumed the same food groups in the last

[11] The score assigns the following weights (in parentheses) to different food groups: Cereals, grains, roots and tubers (2); legumes/ pulses, nuts and seeds (3); milk and other dairy products (4); meat, fish and eggs (4); vegetables and leaves (1); fruits (1); oil, fat and butter (0.5); sugar and sweets (0.5); condiments (0).

24 hours in panel B, weighting responses in the same manner as the household food consumption score with frequency 1 (one day). Both women and men's food consumption in the last 24 hours was significantly higher in response to the loan program. The results suggest that the loan helped finance spending on assets and food security, despite loan repayment being near universal (only 3 out of 353 loan-receiving households defaulted). We expect that the increased spending on productive assets, consumer durables and food security were partially financed by savings from the lower interest payments under this loan program in comparison to other credit alternatives in the market.

Appendix Table A.5 shows that the loan program increased the number of crops by almost 4% and area cultivated by about 13.5% in the treatment arm relative to control[12]. While the treatment arm diversified the number of crops and expanded the area under cultivation, it did not impact returns from farming: real farm income and total quantity harvested did not change over time. The area under cultivation of the main crop (rice, cotton or soybean) did not change in response to the loan program, nor did real farm income or harvest. The loan program did not affect input utilization: households in the treatment villages did not spend any differently from the control on irrigation, purchase of seeds and fertilizers and rent of machinery and land (Table A.6). Table A.7 reports recall of the panel respondent on their own labor allocation as well as the labor supply of male and female household members during the summer agricultural season in 2023. We find no find no significant differences between the treatment and control arms in response to the loan program. The loan also did not affect aggregate wages in the treatment arm nor total expenditure on both male and female labor in the cultivation of the main crops during the summer agricultural season (Table A.8) suggesting that demand for labor did not change with the expansion of total area under cultivation. We further investigate the muted impacts on agricultural intensification by asking households their use of *any formal loans* in the endline survey. Table A.9 shows that loan-receiving households did not apply the loan money any differently than the control group to buying seeds or pesticides and hiring labor or machinery. The treated group households were about 32 pp less likely to use a formal loan for repayment of other loans but this difference is not statistically significant when correcting for false discovery using sharpened q values.

In the next section, we investigate whether the loan changed resource sharing among household members, their financial inclusion, time use and involvement in decision-making.

[12] The sharpened q value that adjusts for multiple hypotheses testing for the point estimates of both total area cultivated and number of crops is 0.109, close to the conventional thresholds.

7 Impact of the loan product on household members

Financial Inclusion

The loan product did not impact women having their own bank account (column 1, Table 6), whereas men in households that received the loan product were 14 percentage points more likely to hold an individual account than men in the control group. This larger effect for men may be influenced by 68% of panel respondents in the treatment arm being men, several of whom may have opened an account in order to receive the loan.

The loan product positively impacted household members' financial autonomy through greater privacy in their transactions. Conditional on individual account ownership, both women and men in households that received the product were more likely to use the debit card linked to their account autonomously (column 2). The treatment effect is significantly larger for women than for men and is driven by differences in the control group mean: fewer than 20% of women in the control arm used their debit card independently, compared with roughly half of the men.

Both women and men also reported greater knowledge and use of mobile banking apps in households that received the loan product relative to the control group (Table 7). For women in loan product-receiving households, this was facilitated by greater smartphone ownership (24 pp) relative to the control group. The intervention did not appear to explicitly change preferences for smartphone ownership as about 74% of households owned at least one smartphone at baseline, and there were no changes on the extensive margin between the two arms over time. This suggests that the loan operated on the intensive margin, enabling women to obtain their own smartphone, that may have been previously outside the household budget. The LATE estimate of women's knowledge of mobile apps exceeds 100 pp, which is possible when the dependent variable is binary. Equivalence tests of the coefficients of knowledge and use of mobile banking apps show that the treatment effects for women are significantly larger than those for men, again owing to a smaller control group mean for women.

The positive impact of the loan product on women's smartphone ownership and uptake of digital financial services is noteworthy, since both mobile ownership and women's internet access have been persistently low in India owing to rigid gender norms (Barboni et al., 2018). Barboni et al. (2024) evaluated a government program distributing smartphones to rural women in Chhattisgarh and found that 40% of the women transferred their phone to another household member and that the program had no impact on women's knowledge or use of mobile banking.

We also find greater resource sharing with women as they were 3.7 pp more likely to deposit funds that were not earnings or transfer payments in their account every week (Table A.10), however, fewer than 2% of the female respondents with an individual bank account answered yes to this question, so the magnitude should be interpreted cautiously.

Time use:
Loan receipt significantly reallocated women's time away from productive activities during the peak agricultural season (Figure 2a). Women in households that received the loan product spent approximately 50% less time on farming activities than women in the control group households (1.58 hours less than the control group mean of 3.13 hours). Reduction of non-agricultural work was more pronounced as women in loan product-receiving households allocated 1.9 hours less than the control group (mean of 1.32 hours). The loan possibly reduced the need for supplementary off-farm income during this time: it led to 24 pp lower probability of women engaging in non-agricultural employment, over 90% less than the control group mean (Table A.11). The decline in time allocation is also linked with household's increased ownership of non-farm business equipment that possibly reduced the demand for female labor. Men's time allocation to non-farm activities was not affected by the loan product. Alongside reductions in work, women in households that received the loan product spent about 0.9 (53%) more hours in personal activities than the control group mean of 1.68 hours, suggesting that time freed from farming and non-agricultural work was redirected toward rest, hobbies or social activities. All three results are statistically significant at conventional thresholds after correcting for multiple hypotheses testing. The loan receipt had no statistically significant effect on women's time spent in livestock and poultry care, consumption activities, or domestic work.

Recent papers on cash transfers show contrasting effects on women's time allocation. In rural Egypt, women reduced their participation in non-farm work but their employment in agriculture did not change in response to the transfer (El-Enbaby et al., 2025). Working for pay in agriculture is socially more acceptable in these regions than non-farm work. Evaluation of cash and food transfers in Ecuador find no effects on women's labor force participation (Hidrobo et al., 2016). Our results also contrast with findings in Takahashi and Barrett (2014) where women reallocated labor from non-farm activities into rice cultivation in response to System of Rice Intensification in Indonesia. In our paper we find that increased liquidity improved household's investment in capital and reduced the demand for female labor.

Men in households that received the loan product spent less time on purchase of large and small items for the household by 0.45 hours and more time on personal activity by 0.37 hours. However, these effects are not statistically significant after correction for multiple hypothesis testing (sharpened q-values of 0.229 for both). The differences in treatment effects of time allocation to farm work and non-agricultural work between women and men are statistically significant (-2.14 and -1.37, respectively).

These patterns are broadly consistent with the results for time use in the day prior to the survey (Figure 2b). For women, loan receipt reduced farming time by 1.44 hours relative to a control group mean of 2.24 hours. Non-agricultural work declined by 0.73 hours relative to a control group mean of 0.95 hours, although the sharpened q-value of this point estimate is just above the threshold of significance (0.146). Women in households that received the loan product also spent 1.01 more hours on personal activity relative to the control group (mean of 1.79 hours). No significant effects are found for men in day prior to the survey, with all sharpened q-values equal to 1. The equality tests between women and men reveal statistically significant gender differences in farming (p = 0.036) and personal activity (p = 0.031), with women reducing farming time by 1.77 hours and increasing personal activity by 0.99 hours more than men.

We find that men's self reports of labor allocation to farming (in Figure 2) is consistent with reports by the panel respondent on labor utilization of male household members (in Table A.7). However, the results of women's self reports differ from the panel respondent's recall of the labor supply of female household members. This is possible because the ITT analysis estimates changes across all households in the treatment villages relative to control in Table A.7 while the 2SLS analysis estimates differences between households that received the product relative to control in Figure 2. Another possible explanation is that most panel respondents were men and did not reliably recall the labor supply of female household members.

Decision-making:

Finally, we test whether improved financial access and time allocations correspond with changes in women's involvement in decision making. Figure 3 reports LATE estimates of the share of decisions women participated in the last 12 months as well as category-wise participation in decision making. The loan product did not affect the number of decisions women or men participated in as a share of total categories tested. It also did not impact women or men's involvement in any decision category on the extensive margin with once exception. Women in the loan product-receiving households were 21 pp less likely participate in decisions on livestock and poultry than the control group (mean = 0.68), however, the sharpened q value of the estimate is 0.507. Comparing aggregate treatment effects between women and men show that women were less likely to be involved in decisions on their personal activities in response to the loan than men (difference = -0.239, p-value=0.09).

The empirical evidence of targeted programs on women's decision making is mixed. Cash transfer programs have had success (de Brauw et al., 2014; Bergolo and Galván, 2018) while social protection programs and bank account ownership did not (Roy et al., 2015; Field et al., 2021; Chauhan, 2025). Women's initial bargaining power in the household explains the differences in program impacts in some contexts (Hidrobo and Fernald, 2013; El-Enbaby et al., 2025). We investigate this mechanism in the next section.

For decisions made jointly with their spouse (Table A.13), women in loan-product receiving households are more likely to participate in livestock and poultry management and household consumption, however, these differences are not statistically significant when corrected for multiple hypothesis testing (sharpened q-value=0.153). There are no differences in men's reports of joint decision making with their spouse between loan product-receiving households and control group. In panel C, we restrict the data to approximate households where spouses were interviewed[13] and find that the the loan product did not differentially impact whether both spouses report joint decision making consistently.

Given differences in the crops planted and production processes in the two states covered under the program, we explore differences in impact of the loan product on household members by state. The positive impacts on financial autonomy (Tables A.14-A.17) and time allocation to personal activities are consistent across both states. Women's time allocation to economic activities in households that received the loan product relative to the control group varied between the two states. Women's labor supply remained unchanged in Odisha in response to the loan product while it declined in Maharahstra. In Odisha, where women's participation in farm and non-agricultural work during peak agricultural season remained unchanged, they were significantly more likely to participate in decisions around both activities than the control group (panel A, Tables A.18 and A.20 respectively). They also allocated more time and were more likely to be involved in household purchases. There is reduced participation in farm work, livestock and poultry and non-agricultural work in Maharashtra (panel B, Tables A.18 and A.20). This also corresponds with lower involvement in decision making on these activities in households that received the loan product relative to control group in Maharashtra (panel B, Table A.22). State-wise results on time allocation are are consistent between peak agricultural season and the day prior to survey except that women in Odisha allocate more time to non-agricultural work.

Men's contribution to farm activities remained unaffected by the loan product in both states, but there were differences between states in men's labor supply to other economic activities during peak agricultural season. In households that received the loan product, men spent significantly less time on livestock and poultry, non agricultural work and household purchase relative to the control group in Mahahrashtra (panel B, Table A.19). Their involvement in decisions around these activities is also significantly lower (panel B, Table A.23). During the peak time, men in Odisha spent relatively more time on

[13] These are households where two members were interviewed successfully in the endline survey. We further assume that the respondents are spouses in cases where either the woman interviewed was upto 19 years younger than the man (95th percentile of the distribution of age difference) or the man was upto 6 years younger than the woman (25th percentile).

non-agricultural work. They allocated more time to personal activities in Odisha in both peak time as well as day prior to survey (panel A, Tables A.19 and A.21 respectively).

8 Mechanisms

In this section, we examine potential factors that could impact individual welfare such as ex-ante measures of household's wealth, socio-economic identity and spousal coordination using heterogeneity tests.

Ex-ante wealth: We construct a wealth index using principal component analysis of the assets owned by a household at baseline and classify all households in our sample by quintile of this index. Interacting this categorical variable with the instrumental variable for whether households received the loan product shows no significant correspondence with women's or men's bank account/ mobile phone ownership and financial autonomy[14] (Tables A.24-A.25). More wealth corresponded with less time spent by women on domestic chores in households that received the product versus control during the peak time in the agricultural season (Table A.26). Men spent less time on livestock and poultry and household purchases, and allocated more time to personal activities. The point estimates of these heterogeneous effects are large, but not statistically significant based on sharpened q-values with the exception for reduced time on household purchases. The heterogeneity in men's time allocation is consistent in the day prior to survey (Table A.27) but there were no heterogeneous effects in women's time allocation to domestic work. There were also no heterogeneous effects on involvement in decision-making (Table A.28).

Ex-ante coordination between spouses: We estimate spousal coordination within a household using a continuous variable of the share of decisions on production and consumption activities that the panel respondent made with their spouse before the loan program. The loan product was effective in increasing women's involvement in decisions in households where there was higher ex-ante coordination between spouses. There was a positive and significant correlation between ex-ante coordination among spouses and women's share of total decisions (Table A.33). Women were more likely to be involved in decisions on livestock and poultry management, non-agricultural work, household consumption, and their own private consumption and personal activities. There were heterogeneous effects on men's involvement in decisions on non-agricultural activities. Spousal coordination in the household corresponded negatively with women's individual account ownership (Table A.29) and positively with mobile phone ownership (Table A.30) but

[14] Men in loan-receiving households were less likely to use their debit cards solely by themselves, but the sharpened q-value of this estimate is greater than conventional levels of significance, and they were no more likely to report sharing the card with family members.

there were no heterogeneous effects on financial autonomy of women or men. Greater spousal coordination encouraged more time allocation to non-agricultural activities and household purchases in loan product-receiving households relative to the control group (Table A.31) during peak time in agriculture. Women also allocated significantly more time to domestic work. In the day prior to survey, there were no heterogeneous effects on men's time allocation while women in households that received the loan product reported more time allocation towards livestock and poultry, non-agriculture and domestic work (Table A.32).

Household's socio-economic identity: Scheduled caste (SC) and scheduled tribes (ST) are historically more disadvantaged, therefore, we investigate if the loan impacted these households differently from the rest (Tables A.34-A.38). There were no heterogeneous effects for SC/ ST households that received the loan product relative to control on financial autonomy indicators. Broadly, time use and involvement in decisions were also not affected by the social identity, except that women spent less time on livestock and poultry, and that men were allocated significantly more time to domestic chores. The trends in time use were consistent both in the peak time of the agricultural season and in the day prior to survey, and women in SC/ST households that received the loan product were more likely to participate in farming decisions than the control group. The sharpened q-values of these point estimates are larger than 0.1.

9 Selection into program

In this section, we distinguish effects on women's agency by household's selection into the loan program. Within the randomly selected treatment villages, households that attended the information session were offered the loan. These 702 households can be further split into 3 categories based on their loan application status - (i) applied and received the loan (352), (ii) applied but were unsuccessful (86) and (iii) did not apply for the loan (310). In Appendix Table A.39, we show that households that received the loan (category i) were statistically similar in observable characteristics with households in the control arm. The unsuccessful loan applicants (category ii) had higher prevalence of literacy and bank account ownership than households in control. However, these differences may be attributed to noise/ high sampling variability from a small sample size of 86. The households that did not apply for the loan (category iii) were less likely to have a bank account or taken a loan the year before the intervention than the control group. Half the households belonged to SC/ST. More than half of them were in the lowest income group captured in the survey ($<$ USD 900 per year) and had lower schooling outcomes (35% of the panel respondents had completed secondary schooling vs. 50% in control).

Given these systematic differences within groups in treatment, we follow Crépon et al. (2015) and estimate a propensity score of borrowing based on observable characteristics in the baseline survey (Figure A.3). We re-weight the LATE by each household's ex-ante predicted probability of loan receipt in Tables A.40 - A.44. The positive impacts on women and men's financial autonomy, greater time allocation to personal activities and no aggregate impacts on involvement in decision-making are consistent in the re-weighted estimation. The effect sizes in section 7 are larger than the re-weighted estimates suggesting that households receiving the loan product experienced larger effects than the average likely borrower would. There are some important differences in outcomes as well: men were more likely to share their debit card with other family members, and women and men's time allocation across different activities varied in some of the economic activities. During peak agricultural season, women spent less time all economic activities including livestock and poultry management. Men also reduced their time on livestock and poultry and non-agricultural work and spent more time on domestic chores.

10 Conclusion

An important question for effective delivery of social protection and financial inclusion is whether they should target the household or individuals. Programs transferring resources to women may improve household profits but don't usually succeed in improving women's income relative to household members or their say in household decisions. Evaluations of loan programs to female entrepreneurs and cash transfers find evidence of resource appropriation and male backlash in several contexts highlighting the potential risks to women through these targeted interventions. In this paper, we investigate an alternative program that indirectly improves women's welfare through smartphone ownership, financial autonomy and leisure.

Agricultural loan subsidies or collateral-free credit are provided by governments to encourage agricultural investment among small and marginal farmers in developing countries. Empirical evidence from these programs focuses on intensification practices and labor supply response to the loan money. The impact on household welfare and particularly individual gains have been under studied. Given that agricultural production is organized at the household level in India, we anticipate that household's utilization of credit may not vary based on which individual is assigned as beneficiary. Instead, individual members may benefit from the additional resources in two ways. First, the loan raises the household's joint income in the short term and directly increases the consumption of composite goods such as productive capital and consumer durables that are shared or privately consumed. Second, individual members may adjust their private consumption and labor allocation in response to household's consumption of composite goods and their own residual income.

In this setting, we analyze the impact of agricultural credit on women's outcomes in India using an RCT where a collateral-free loan was provided to small and marginal farmers in 40 of the 80 study villages. We combine the randomized loan provision with a balanced panel of 1,429 households interviewed in baseline and endline surveys. Households in both treatment and control groups were balanced on observable characteristics at baseline. We analyze the intent-to-treat effects of the loan program on household resource allocation using an ANCOVA model. We also collect individual level outcomes in the endline survey, and use a 2SLS model to estimate the local average treatment effect of the loan on individual financial inclusion and time use. This model instruments the exogenous village-level treatment assignment for households receiving the loan product.

There are three key results: first, households increased ownership of productive capital for farm work (14 pp) and business (7 pp) in response to the loan program. They also increased ownership of large consumer durables (8 pp) and food security improves (17%). The area under cultivation increased but there were no changes in spending on inputs or income from agriculture. Conditional on individual bank account, women in loan product-receiving households reported greater knowledge (LATE estimates of the binary outcome variable is over 100 pp) and use of mobile banking services (94 pp). The significantly greater uptake of digital financial services in the treatment arm was driven by smartphone ownership (24 pp) and women's preference for financial autonomy: They were 81 pp more likely to use their debit card alone. Third, increased resources shift household members' labor supply. Women in treatment spend less time on farm work (1.6 vs. 3.1 hours) and more on leisure (2.6 vs. 1.7 hours). In the absence of income changes, it seems that the equipment purchased reduces the requirements for women's labor and households did not hire more labor in response to the program. Women's participation in non-agricultural economic activities reduced as well suggesting less demand for non-farm income in response to the loan. There were no impacts of the loan product on women's involvement in decision making on different production and non-production activities, however, in households with greater ex-ante coordination among spouses on decisions, they were more involved in decision-making.

For policymakers, this evidence highlights the downstream welfare effects of agricultural support programs that aren't accounted for when evaluating their benefits. Women in rural agrarian households typically face limited financial access (Demirguc-Kunt et al., 2022). By positively impacting women's independent use of financial services, such interventions extend benefits beyond farm productivity and income.

CRediT authorship contribution statement

Tarana Chauhan: Writing – original draft, Writing – review & editing, Conceptualization, Methodology, Software, Formal analysis, Visualization. **Berber Kramer**: Conceptualization, Funding acquisition, Methodology, Review & Editing. **Patrick Ward**: Conceptualization, Funding acquisition, Methodology. **Subhransu Pattnaik**: Software, Data curation.

Declaration of Competing Interest

The authors have no relevant or material financial interests that relate to the research described in this paper.

References

Afridi, F., Bishnu, M., and Mahajan, K. (2023). Gender and mechanization: Evidence from Indian agriculture. *American Journal of Agricultural Economics*, 105(1):52–75. 2

Allendorf, K. (2007). Do Women's Land Rights Promote Empowerment and Child Health in Nepal? *World Development*, 35(11):1975–1988. 14

Angelucci, M. (2008). Love on the Rocks: Domestic Violence and Alcohol Abuse in Rural Mexico. *B.E. Journal of Economic Analysis & Policy*, 8(1). 2

Angelucci, M., Karlan, D., and Zinman, J. (2015). Microcredit Impacts: Evidence from a Randomized Microcredit Program Placement Experiment by Compartamos Banco. *American Economic Journal: Applied Economics*, 7(1):151–82. 2, 5

Annan, J., Donald, A., Goldstein, M., Martinez, P. G., and Koolwal, G. (2021). Taking power: Women's empowerment and household Well-being in Sub-Saharan Africa. *World Development*, 140:105292. 14

Attanasio, O., Augsburg, B., De Haas, R., Fitzsimons, E., and Harmgart, H. (2015). The Impacts of Microfinance: Evidence from Joint-Liability Lending in Mongolia. *American Economic Journal: Applied Economics*, 7(1):90–122. 2

Banerjee, A., Duflo, E., Glennerster, R., and Kinnan, C. (2015). The Miracle of Microfinance? Evidence from a Randomized Evaluation. *American Economic Journal: Applied Economics*, 7(1):22–53. 2, 5

Barboni, G., Bhattacharya, A., Field, E., Pande, R., Rigol, N., Schaner, S., Shukla, A., and Moore, C. T. (2024). Hold the Phone: The Short- and Long-Run Impacts of Connecting Indian Women to Digital Technology. *EGC Discussion Paper*. 4, 19

Barboni, G., Field, E., Pande, R., Rigol, N., Schaner, S., and Moore, C. T. (2018). A Tough Call: Understanding barriers to and impacts of women's mobile phone adoption in India. Technical report, Harvard Kennedy School Evidence for Policy Design. 4, 19

Barooah, P., Birner, R., Bosch, C., Alvi, M., Munshi, S., Kumar, A., and Kumar, V. (2025). Sowing Change: What Mechanized Direct Seeded Rice (DSR) Reveals about Equity in Climate-Smart Transitions? Blog post, IFPRI South Asia Office. Accessed: May 18, 2026. 10

Beaman, L., Karlan, D., Thuysbaert, B., and Udry, C. (2023). Selection Into Credit Markets: Evidence From Agriculture in Mali. *Econometrica*, 91(5):1595–1627. 5

Belloni, A., Chernozhukov, V., and Hansen, C. (2013). Inference on Treatment Effects after Selection among High-Dimensional Controls. *The Review of Economic Studies*, 81(2):608–650. 15

Bergolo, M. and Galván, E. (2018). Intra-household Behavioral Responses to Cash Transfer Programs. Evidence from a Regression Discontinuity Design. *World Development*, 103:100–118. 21

Bernhardt, A., Field, E., Pande, R., and Rigol, N. (2019). Household Matters: Revisiting the Returns to Capital among Female Microentrepreneurs. *American Economic Review: Insights*, 1(2):141–60. 2, 5

Bobonis, G. J. (2009). Is the Allocation of Resources within the Household Efficient? New Evidence from a Randomized Experiment. *Journal of Political Economy*, 117(3):453–503. 6, 7

Bobonis, G. J., Gonzalez-Brenes, M., and Castro, R. (2013). Public transfers and domestic violence: The roles of private information and spousal control. *American Economic Journal: Economic Policy*, 5(1). 2

Browning, M. and Chiappori, P.-A. (1998). Efficient Intra-Household Allocations: A General Characterization and Empirical Tests. *Econometrica*, 66(6):1241–1278. 7

CGAP (2019). India: A Testing Ground for Digital Merchant Payments. Technical report, CGAP. 14

Chauhan, T. (2025). Accounting for Empowerment? Examining Women's Financial Inclusion in India. *GLO Discussion Paper No. 1689*. 21

Chiappori, P.-A. (1988). Rational Household Labor Supply. *Econometrica*, 56(1):63–90. 6

Connelly, R., Roberts, K., and Zheng, Z. (2010). The Impact of Circular Migration on the Position of Married Women in Rural China. *Feminist Economics*, 16(1):3–41. 14

Crépon, B., Devoto, F., Duflo, E., and Parienté, W. (2015). Estimating the Impact of Microcredit on Those Who Take It Up: Evidence from a Randomized Experiment in Morocco. *American Economic Journal: Applied Economics*, 7(1):123–50. 25

de Brauw, A., Gilligan, D. O., Hoddinott, J., and Roy, S. (2014). The Impact of Bolsa Família on Women's Decision-Making Power. *World Development*, 59:487–504. 21

de Mel, S., McKenzie, D., and Woodruff, C. (2008). Returns to capital in microenterprises: Evidence from a field experiment. *The Quarterly Journal of Economics*, 123(4):1329–1372. 5

Demirguc-Kunt, A., Klapper, L., Singer, D., and Ansar, S. (2022). The Global Findex Database 2021 : Financial Inclusion, Digital Payments, and Resilience in the Age of COVID-19 - Chapter 2 : Use of Financial Services. Technical report, World Bank Group. 26

Doss, C. R. (2018). Women and agricultural productivity: Reframing the Issues. *Development Policy Review*, 36(1):35–50. 5

Doss, C. R., Meinzen-Dick, R., Pereira, A., and Pradhand, R. (2022). Women's empowerment, extended families and male migration in Nepal: Insights from mixed methods analysis. *Journal of Rural Studies*, 90:13–25. 14

Dwivedi, M. and Damle, N. (2019). Business Case for Gender Mainstreaming in Cotton in Maharashtra. Report, IDH The Sustainable Trade Initiative and Sattva Research, Amsterdam, Netherlands. Commissioned by IDH The Sustainable Trade Initiative. Accessed: May 18, 2026. 10

El-Enbaby, H., Gilligan, D. O., Karachiwalla, N., Kassim, Y., and Kurdi, S. (2025). Cash Transfers, Gender Norms, and Women's Control over Decision-Making in Egypt. *Economic Development and Cultural Change*, 73(4):1721–1760. 5, 14, 20, 21

Fiala, N. (2017). Business is tough, but family is worse: Household bargaining and investment in microenterprises in Uganda. Working paper. 2

Fiala, N. (2018). Returns to microcredit, cash grants and training for male and female microentrepreneurs in Uganda. *World Development*, 105:189–200. 2, 5

Field, E., Pande, R., Rigol, N., Schaner, S., and Moore, C. T. (2021). On Her Account: How Strengthening Women's Financial Control Impacts Labor Supply and Gender Norms. *American Economic Review*, 111(7):2342–75. 5, 14, 21

Fink, G., Jack, B. K., and Masiye, F. (2020). Seasonal Liquidity, Rural Labor Markets, and Agricultural Production. *American Economic Review*, 110(11):3351–92. 5, 14

Friedson-Ridenour, S. and Pierotti, R. S. (2019). Competing priorities: Women's microenterprises and household relationships. *World Development*, 121:53–62. 5

Garikipati, S. (2008). The Impact of Lending to Women on Household Vulnerability and Women's Empowerment: Evidence from India. *World Development*, 36(12):2620–2642. 5

Green, E. P., Blattman, C., Jamison, J., and Annan, J. (2015). Women's entrepreneurship and intimate partner violence: A cluster randomized trial of microenterprise assistance and partner participation in post-conflict Uganda. *Social Science & Medicine*, 133:177–188. 2

Gulati, K., Saha, K., and Lybbert, T. J. (2025). Women's work and agricultural productivity gaps in India. *American Journal of Agricultural Economics*, 107(5):1261–1289. 2

Haushofer, J. and Shapiro, J. (2016). The Short-term Impact of Unconditional Cash Transfers to the Poor: Experimental Evidence from Kenya. *The Quarterly Journal of Economics*, 131(4):1973–2042. 3

Heckert, J., Martinez, E. M., Sanou, A., Pedehombga, A., Ganaba, R., and Gelli, A. (2023). Can a gender-sensitive integrated poultry value chain and nutrition intervention increase women's empowerment among the rural poor in Burkina Faso? *Journal of Rural Studies*, 100:103026. 14

Hidrobo, M. and Fernald, L. (2013). Cash transfers and domestic violence. *Journal of Health Economics*, 32(1):304–319. 21

Hidrobo, M., Peterman, A., and Heise, L. (2016). The Effect of Cash, Vouchers, and Food Transfers on Intimate Partner Violence: Evidence from a Randomized Experiment in Northern Ecuador. *American Economic Journal: Applied Economics*, 8(3):284–303. 5, 14, 20

Karlan, D., Osei, R., Osei-Akoto, I., and Udry, C. (2014). Agricultural Decisions after Relaxing Credit and Risk Constraints . *The Quarterly Journal of Economics*, 129(2):597–652. 5

Karlan, D., Savonitto, B., Thuysbaert, B., and Udry, C. (2017). Impact of savings groups on the lives of the poor. *Proceedings of the National Academy of Sciences*, 114:3079–3084. 5

Kochar, A., Nagabhushana, C., Sarkar, R., Shah, R., and Singh, G. (2022). Financial access and women's role in household decisions: Empirical evidence from India's National Rural Livelihoods project. *Journal of Development Economics*, 155:102821. 14

Kosec, K., Akramov, K., Mirkasimov, B., Song, J., and Zhao, H. (2022). Aspirations and women's empowerment: Evidence from Kyrgyzstan. *Economics of Transition and Institutional Change*, 30(1):101–134. 14

Kramer, B., Pattnaik, S., Ward, P. S., and Xu, Y. (2025). Bundling crop insurance with digital credit to boost agricultural investments: Findings from a randomized trial in two states in India. *Working Paper*. 3, 5, 13, 15, 16, 47

Maitra, P., Mitra, S., Mookherjee, D., Motta, A., and Visaria, S. (2017). Financing smallholder agriculture: An experiment with agent-intermediated microloans in India. *Journal of Development Economics*, 127:306–337. 5

Mukherjee, S. W., Bergquist, L. F., Burke, M., and Miguel, E. (2024). Unlocking the benefits of credit through saving. *Journal of Development Economics*, 171:103346. 5

Nakano, Y. and Magezi, E. F. (2020). The impact of microcredit on agricultural technology adoption and productivity: Evidence from randomized control trial in Tanzania. *World Development*, 133:104997. 5

Quisumbing, A., Cole, Steven andElias, M., Faas, S., Galiè, A., Malapit, H., Meinzen-Dick, R., Myers, E., Seymour, G., and Twyman, J. (2023a). Measuring Women's Empowerment in Agriculture: Innovations and evidence. *Global Food Security*, 38(100707):1–15. 14

Quisumbing, A., Meinzen-Dick, R., and Malapit, H. (2023b). Measuring women's empowerment and gender equality through the lens of induced innovation. In *Agricultural Development in Asia and Africa: Essays in Honor of Keijiro Otsuka*, pages 343–355. Springer Nature Singapore. 14

Riley, E. (2024). Resisting Social Pressure in the Household Using Mobile Money: Experimental Evidence on Microenterprise Investment in Uganda. *American Economic Review*, 114(5):1415–47. 2, 5

Roy, S., Ara, J., Das, N., and Quisumbing, A. R. (2015). "Flypaper effects" in transfers targeted to women: Evidence from BRAC's "Targeting the Ultra Poor" program in Bangladesh. *Journal of Development Economics*, 117:1–19. 21

Said, F., Mahmud, M., d'Adda, G., and Chaudhr, A. (2019). Home-bias among Female Entrepreneurs: Experimental Evidence on Preferences from Pakistan. CREB Working Paper No. 04-20. 2

Samuelson, P. A. (1956). Social Indifference Curves. *The Quarterly Journal of Economics*, 70(1):1–22. 6

Schaner, S. (2016). The Cost of Convenience? Transaction Costs, Bargaining Power, and Savings Account Use in Kenya. *Journal of Human Resources*, 52(4):919–945. 14

Swaminathan, H., Lahoti, R., and Suchita, J. (2012). Women's Property, Mobility, and Decisionmaking: Evidence from Rural Karnataka, India. *IFPRI Discussion Paper*, 1188. 14

Takahashi, K. and Barrett, C. B. (2014). The System of Rice Intensification and its Impacts on Household Income and Child Schooling: Evidence from Rural Indonesia. *American Journal of Agricultural Economics*, 96(1):269–289. 20

Tarozzi, A., Desai, J., and Johnson, K. (2015). The Impacts of Microcredit: Evidence from Ethiopia. *American Economic Journal: Applied Economics*, 7(1):54–89. 5

11 Figure

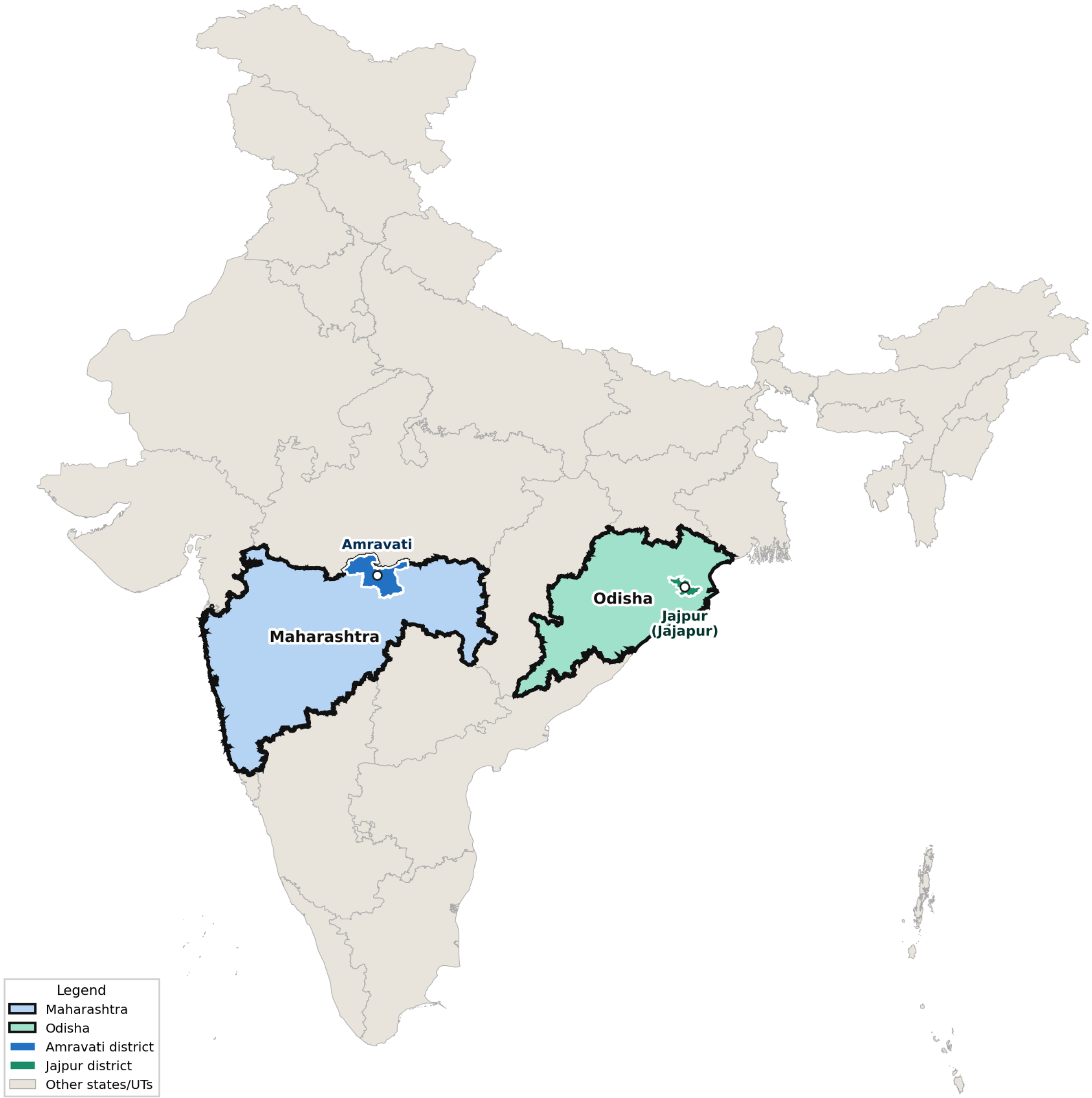
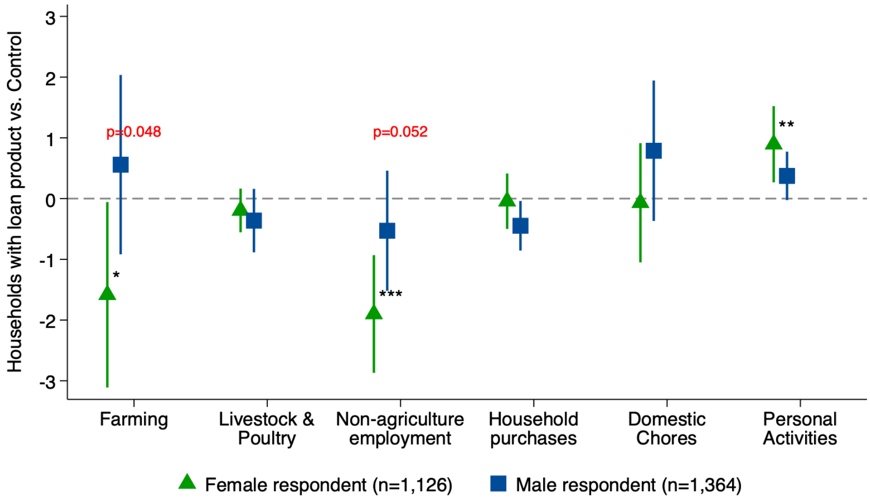
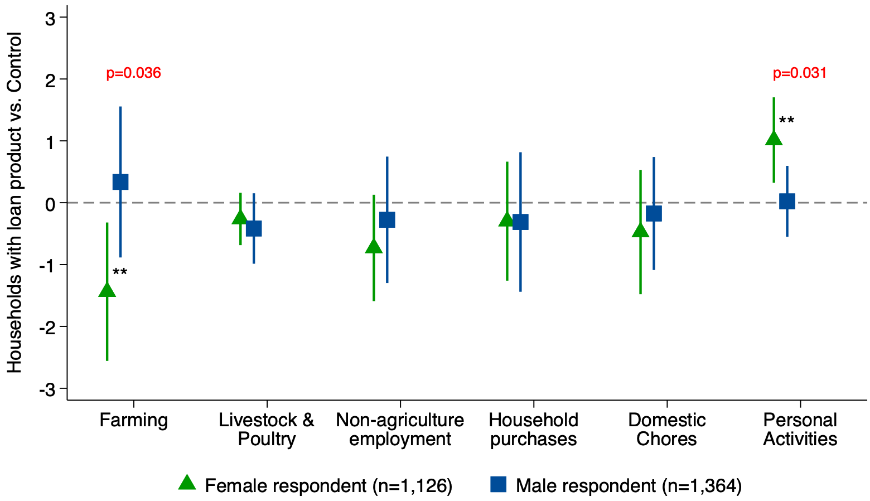


Figure 1: Study location

Notes: The figure indicates the two districts in India where the program was implemented - Amravati in Mahrahstra (blue) and Jajpur in Odisha (green).



(a) Peak time in agricultural season



(b) Day prior to survey (yesterday)

Figure 2: LATE: Loan intervention and individual time use

Notes: The figure the local average treatment effect of the loan on women and men's time use. Panel A reports time spent on production and non-production activities (described in Table 2) in the peak time of the agricultural season and panel B reports time allocation on the day prior to the survey. Sharpened q values report statistical significance of coefficients after correcting for multiple hypothesis testing. $*$ q$<$ 0.1, $**$ q$<$ .05, $***$ q$<$ .01. The p-value of an equivalence test is reported if the coefficients of men and women differ significantly from each other.

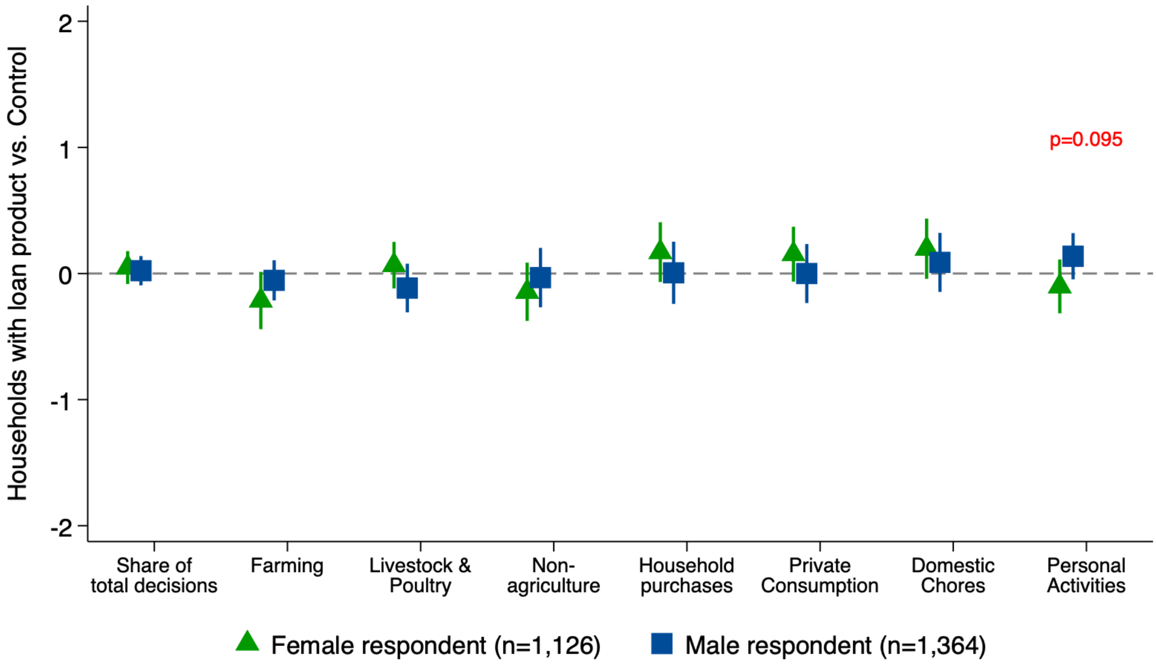


Figure 3: LATE: Loan intervention and involvement in decision-making

Notes: The figure the local average treatment effect of the loan on women and men's involvement in decision making on production and non-production activities (described in Table 2). Sharpened q values report statistical significance of coefficients after correcting for multiple hypothesis testing. $*$ q$<$ 0.1, $**$ q$<$ .05, $***$ q$<$ .01. The p-value of an equivalence test is reported if the coefficients of men and women differ significantly from each other.

12 Tables

Table 1: Baseline covariates by intervention arm

	(1) All	(2) Loan	(3) Control	F-test for balance across all groups	(2)-(3) Difference
Respondent's age	43.301	43.824	42.727	1.639	1.097
	(-0.447)	(-0.723)	(-0.469)	0.204	
Respondent is female	0.367	0.37	0.363	0.008	0.008
	(-0.042)	(-0.052)	(-0.067)	0.929	
Respondent is married	0.934	0.941	0.927	0.484	0.015
	(-0.01)	(-0.011)	(-0.018)	0.489	
Can read and write	0.875	0.872	0.88	0.065	-0.008
	(-0.016)	(-0.021)	(-0.023)	0.8	
Completed at least primary school	0.788	0.769	0.809	0.889	-0.04
	(-0.022)	(-0.03)	(-0.031)	0.349	
Completed at least	0.457	0.413	0.505	2.516	-0.092
secondary school	(-0.029)	(-0.039)	(-0.044)	0.117	
Respondent belongs	0.358	0.39	0.323	0.991	0.067
to SC/ST category	(-0.035)	(-0.05)	(-0.046)	0.322	
Annual Household	0.474	0.511	0.433	0.82	0.078
Income(<90,000)	(-0.043)	(-0.059)	(-0.063)	0.368	
Annual Household Income	0.237	0.235	0.239	0.008	-0.004
(90,000 - 120,000)	(-0.023)	(-0.032)	(-0.034)	0.93	
Annual Household	0.289	0.254	0.327	0.978	-0.073
Income(>120,000)	(-0.038)	(-0.053)	(-0.053)	0.326	
Owns land	0.479	0.493	0.464	0.16	0.029
	(-0.037)	(-0.05)	(-0.054)	0.69	
Asset quintile	2.986	3.043	2.924	0.273	0.119
	(-0.114)	(-0.155)	(-0.169)	0.602	
Borrowed money last	0.439	0.418	0.463	0.428	-0.044
12 months	(-0.034)	(-0.046)	(-0.05)	0.515	
Has outstanding loan	0.212	0.225	0.198	0.141	0.026
	(-0.035)	(-0.051)	(-0.048)	0.708	
Respondent has bank	0.933	0.929	0.937	0.158	-0.008
account	(-0.01)	(-0.013)	(-0.014)	0.692	
F-test of joint significance (F-stat)				0.725	
N	1429	748	681		

Notes: The table reports averages of respondent and household characteristics from the baseline survey by intervention arm. Column 1 reports pooled estimates for the full sample. Column 2 reports the averages for the loan treatment arm and column 3 for control arm. The fourth column reports the f test for whether each variable is balanced across arms and the last column conducts a pairwise t-test of the mean difference between columns 2 and 3. “SC/ST" is abbreviation for Scheduled Caste/ Scheduled Tribe. Standard errors are clustered at the level of treatment assignment (village) and reported below mean values for each variable. The F-test of joint significance omits the highest household income category to test for variance across all variables. The number of observations are included in the last row. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 2: Variable description of activities

Variable	Description
Farming	- Farming and processing of the harvest: grains that are grown primarily for food consumption (rice, maize, wheat, millet) - Horticultural (gardens) or high value crop farming and processing of the harvest
Livestock & Poultry	- Raising cattle, buffaloes, sheep, goats, pigs and processing of their milk and/or meat; raising chickens, ducks, turkeys and processing their eggs and/or meat
Non-agriculture	- Small business, self-employment, buy-and-sell - Wage and salary employment
Household purchases	- Large, occasional household purchases (bicycles, land, transport vehicles) - Routine household purchases (food for daily consumption or other household needs)
Domestic work	Cooking, cleaning, fetching firewood or water, and caring for children or others
Personal activity	Eating, sleeping, leisure such as visiting neighbors, watching TV, listening to the radio, seeing movies, or doing sports

Notes: Variable generation for analyzing time use and decision making from individual activities asked in the survey.

Table 3: ITT: Loan intervention and household asset ownership

	Non-mechanized farm equipment	Mechanized farm equipment	Business equipment	House and structures	Large consumer durables	Small consumer durables
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Loan	0.138**	-0.001	0.071**	-0.034	0.078***	-0.071
	(0.067)	(0.041)	(0.029)	(0.026)	(0.029)	(0.057)
Sharpened q-values	0.061	0.345	0.052	0.147	0.052	0.147
Observations	1316	1305	1294	1318	1308	1301
Mean dependent variable (Baseline)	0.241	0.275	0.132	0.742	0.812	0.527
Covariates included	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on household ownership of assets (listed as column titles). The dependent variables are binary and constructed from survey questions on whether household owns the following: Column 1 includes non-mechanized tools for farming such as hand tools, animal-drawn plows and non-mechanized irrigation tools. Column 2 includes mechanized farm equipment such as electric pump-set, diesel pump-set, tractor, seeder, power-tiller, thresher, combine harvester etc. Column 3 includes non-farm business equipment such as sewing machines and solar panels. Column 4 includes large consumer durables such as refrigerator, TV, sofa, dining table, chair, mattress, cot or bed, computer, washing machine. Column 5 includes small consumer durables (radio or transistor, pressure cooker, an inverter, electric fan, watch or clock, landline phone). The model includes outcome variable and covariates from the baseline survey as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ $p < 0.1$, $**$ $p < .05$, $***$ $p < .01$

Table 4: ITT: Food Consumption Score (FCS)

	(1)	
Panel A: Household food consumption score		
ITT estimate	5.611*	
	(2.845)	
Observations	701	
Mean dependent variable (Baseline)	33.799	
Covariates included	Yes	
Block fixed effects	Yes	
Panel B: Individual food consumption score (24 hrs recall)		
	Women	Men
	(1)	(2)
LATE estimate	1.966*	2.829*
	(1.183)	(1.552)
Observations	395	342
Control group mean	10.661	10.661
Covariates included	Yes	Yes
Block fixed effects	Yes	Yes

Notes: The table reports the effect of loan treatment on food consumption score in panel A (ITT estimates), and the effect of household receiving the loan product on individual food consumption score in last 24 hours in panel B (LATE estimates). Panel A uses an ANCOVA model where outcome variables and covariates measured at baseline are used as regressors in the estimation. Panel B uses treatment assignment as an exogenous instrument for the household's receipt of loan product. All models control for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 5: Instrumental Variables: First stage

	(1)	(2)	(3)
Treatment: Loan	0.471***	0.472***	0.481***
	(0.046)	(0.043)	(0.039)
Constant	0.000	0.006	0.245
	(0.000)	(0.103)	(0.172)
Observations	1429	1429	1429
Baseline covariates	No	Yes	Yes
Block fixed effects	No	No	Yes
F value	105.939	18.128	18.616
Partial R-squared		0.294	0.303

Notes: The table reports ordinary least squares estimation of village's assignment to treatment on household receiving loan product for summer cultivation in 2023. This estimation uses the endline survey only. Columns 2 and 3 include controls measured at baseline and described in Table 1. Column 3 includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 6: LATE: Individual bank account and debit card

	Own bank account (1)	Use debit card alone (2)	Share debit card (3)
	Panel A: Women		
Received loan	-0.021	0.813***	-0.110
product (γ_F)	(0.096)	(0.122)	(0.104)
Constant	0.460**	0.736***	0.069
	(0.216)	(0.179)	(0.205)
Sharpened q-values	0.766	0.001	0.407
Observations	1126	968	968
Control group mean	0.857	0.183	0.318
	Panel B: Men		
Received loan	0.140*	0.307***	0.019
product (γ_M)	(0.075)	(0.090)	(0.091)
Constant	0.643***	1.236***	-0.099
	(0.148)	(0.179)	(0.133)
Sharpened q-values	0.065	0.004	0.384
Observations	1364	1207	1207
Control group mean	0.836	0.524	0.236
Difference ($\gamma_F - \gamma_M$)	-0.162	0.506	-0.130
p-value of test ($\gamma_F = \gamma_M$)	0.183	0.001	0.347
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on household members' bank account ownership and use of debit card. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Columns 2 and 3 are conditional on respondent owning an individual bank account. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 7: LATE: Mobile phone and digital financial services

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
Panel A: Women				
Received loan	0.248*	0.240**	1.034***	0.940***
product (γ_F)	(0.130)	(0.116)	(0.117)	(0.104)
Sharpened q-values	0.029	0.027	0.001	0.001
Observations	625	663	968	968
Control group mean	0.322	0.287	0.213	0.173
Panel B: Men				
Received loan	-0.100	0.044	0.495***	0.439***
product (γ_M)	(0.124)	(0.113)	(0.081)	(0.081)
Constant	0.195	0.969***	1.318***	1.380***
	(0.162)	(0.177)	(0.163)	(0.151)
Sharpened q-values	0.388	0.533	0.001	0.001
Observations	732	874	1207	1207
Control group mean	0.629	0.730	0.610	0.579
Difference ($\gamma_F - \gamma_M$)	0.348	0.195	0.539	0.501
p-value of test ($\gamma_F = \gamma_M$)	0.053	0.227	0.000	0.000
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on household members' smartphone ownership and uptake of digital financial services. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. The analysis is conditional on respondent owning an individual bank account. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

13 Appendix Figure

List of Appendix Figures

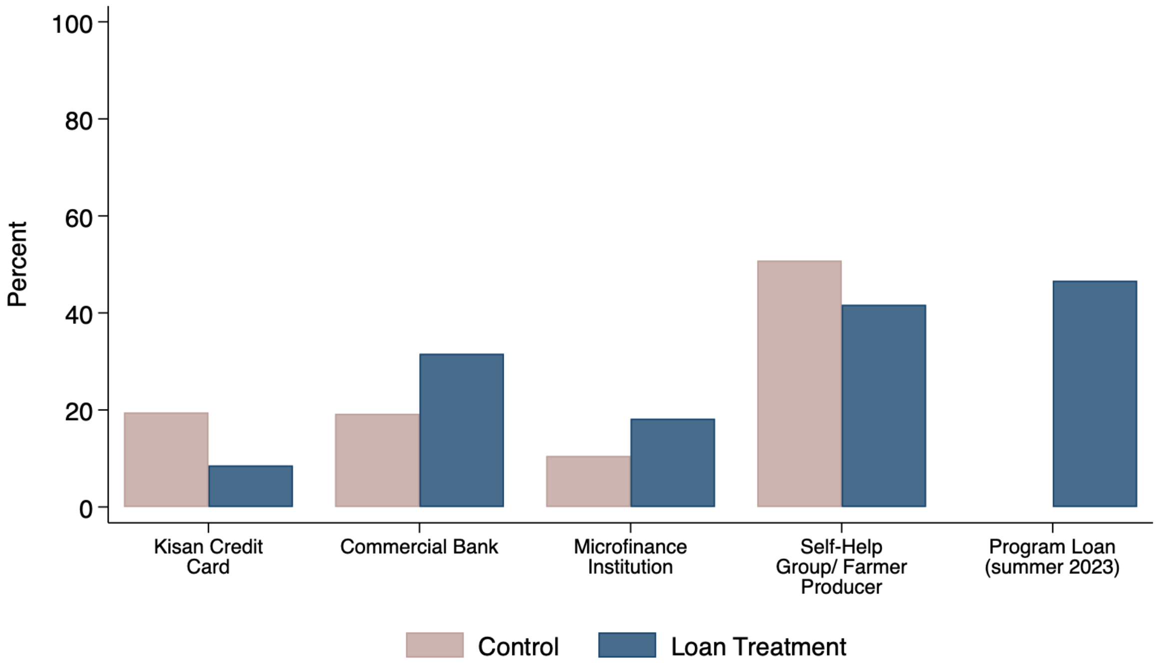


Figure A.1: Sources of formal loan by intervention arm

Notes: The figure plots the distribution of uptake of formal sources of credit for the summer planting season in 2023 reported by households in the endline survey.

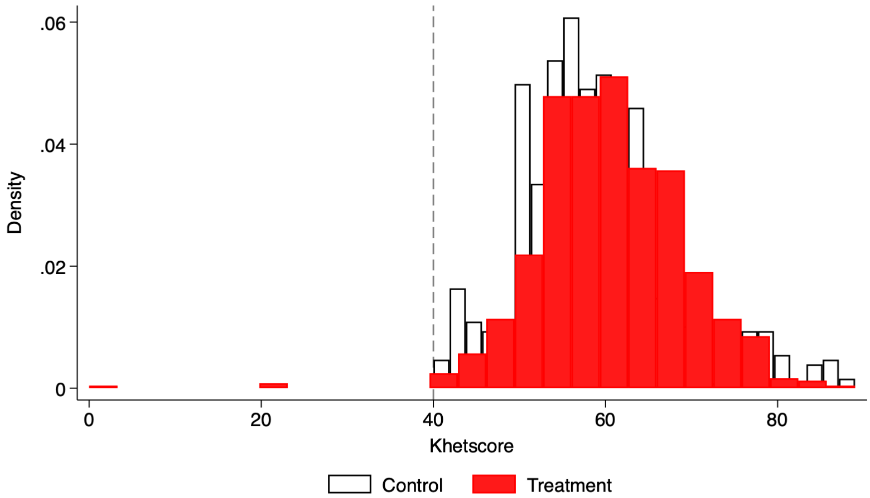


Figure A.2: Distribution of sample by credit score

Notes: The figure plots the distribution of credit score of households in the control (black outline) and treatment arm (solid red). The score is a weighted index computed by DER and the loan eligibility threshold was 40.

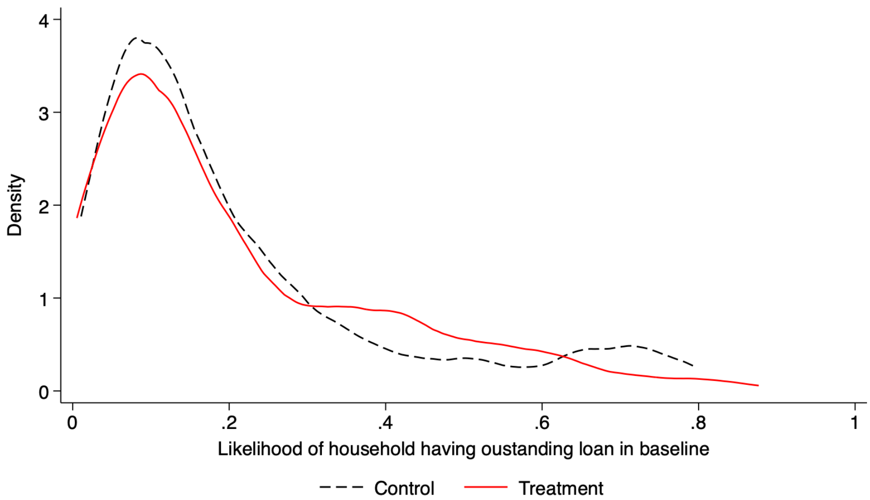


Figure A.3: Distribution of propensity to borrow by intervention arm

Notes: The figure plots the distribution of propensity score of household receiving a loan at baseline in the control (black) and treatment arm (red). The score is estimated using a logit model where the outcome is a binary variable of whether the household had an outstanding loan and household characteristics from the baseline survey are explanatory variables.

14 Appendix Tables

List of Appendix Tables

Table A.1: Cost of the loan program and net returns

Fixed costs (per season)		
Project management	8733.33	
Staff travel	1071.33	
Field office cost	4000.00	
App/portal maintenance	5158.67	
Communication material	178.67	
Total fixed cost per season	19142.00	
Total fixed cost per farmer	30.58	
	Maharashtra	Odisha
Variable costs (per farmer)		
Khetscore report	5.5	5.5
DER field staff	10.56	10.56
Staff transport, allowance	2.079	2.156
Printing, stationary	0.352	0.363
Total Variable cost per farmer	18.491	18.579
Interest 14% of loan	72.38	63.14
Number of beneficiaries	160	193
Defaulters	0	3
Total costs= (fixed + variable)	7851.36	9487.69
Total revenue	11580.80	11996.60
Net return	3729.44	2508.91
Net return per farmer	23.31	13.00

Notes: The cost of the program (in USD) is compiled using data shared by DER. Total cost per farmer is estimated using total farmers who received the loan product across the entire intervention which included two treatment arms. This paper only analyzes the loan treatment arm while Kramer et al. (2025) documents the additional loan and insurance treatment.

Table A.2: Attrition by treatment assignment and baseline characteristics

	Sample that attrited	
	(1)	(2)
Treatment: Loan	-0.038	-0.033
	(0.036)	(0.037)
Respondents' age		0.001
		(0.002)
Respondent is female		0.061**
		(0.025)
Can read and write		0.027
		(0.029)
Completed at least primary school		-0.008
		(0.027)
Completed at least secondary school		-0.013
		(0.025)
Respondent belongs to SC/ST category		-0.023
		(0.022)
Respondent is married		0.021
		(0.031)
Annual Household Income(<90,000)		0.024
		(0.029)
Annual Household Income (> 120,000)		0.024
		(0.024)
Owns land		-0.004
		(0.024)
Respondent has bank account		-0.044
		(0.044)
Borrowed money last 12 months		0.018
		(0.024)
Have outstanding loan		-0.059*
		(0.033)
Constant	0.152***	0.134
	(0.023)	(0.083)
Observations	1682	1682
Non-attrited (N)	1431	1431
Block dummies	Yes	Yes
Mean dep. variable	0.149	

Notes: The table reports reduced form estimation of whether attrition of the sample was significantly correlated with treatment assignment (column 1) or ex-ante characteristics of the household measured from the baseline survey (column 2). The binary dependent variable distinguishes the sample that was interviewed at baseline and not in the endline survey. Both models include block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. The average sample that attrited in the control group is reported in the last row of column 1. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.3: Administrative data on loan take-up

	Summer 2023	
	Control	Loan Treatment
	(1)	(2)
DER generated Credit score	58.9	60.5
Applied for loan	0	58.5%
Received loan	0	47%
Loan amount (INR)	0	41, 505
Default		0.85%
Observations	681	748

Notes: The table reports the disbursal of the program's loan product to households in the control and treatment arm. Column 1 verifies the validity of the experiment design as no control group households were given the loan. This data is extracted from the loan records of the implementing partner DER.

Table A.4: Balance test of outcome variables at baseline

		(1)		(2)		(3)	F-test for balance	(2)-(3)
		All		Loan		Control	across all groups	Pairwise t-test
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	F-stat/P-value	Mean difference
Income	1319	1071.018	701	1090.449	618	1048.979	0.01	41.47
	80	(-208.991)	40	(-351.899)	40	(-206.698)	(0.919)	
Income main crop	928	1202.374	487	1057.025	441	1362.884	1.732	-305.859
	78	(-120.477)	39	(-167.085)	39	(-163.539)	(0.192)	
Harvest (quintiles)	1429	35.424	748	36.93	681	33.769	0.902	3.162
	80	(-1.724)	40	(-2.564)	40	(-2.155)	(0.345)	
Harvest main crop (quintiles)	850	30.595	415	29.075	435	32.045	0.884	-2.97
	68	(-1.586)	35	(-2.178)	33	(-2.319)	(0.351)	
Yield (Harvest/ Area)	1361	14.394	722	15.336	639	13.33	0.895	2.006
	80	(-1.078)	40	(-1.46)	40	(-1.556)	(0.347)	
Yield (Harvest/ Area) main crop	850	10.01	415	10.176	435	9.852	0.041	0.325
	68	(-0.807)	35	(-1.13)	33	(-1.163)	(0.841)	
Total area (acres)	1429	2.966	748	2.773	681	3.179	1.168	-0.406
	80	(-0.185)	40	(-0.194)	40	(-0.325)	(0.283)	
Total area (acres) main crop	1325	3.049	705	2.771	620	3.366	3.215*	-0.595*
	80	(-0.165)	40	(-0.166)	40	(-0.29)	(0.077)	
Number crops	1429	2.373	748	2.412	681	2.33	1.519	0.081
	80	(-0.034)	40	(-0.051)	40	(-0.042)	(0.221)	
Expenditure male labor	1067	27.125	563	37.863	504	15.129	7.245***	22.734***
	80	(-4.778)	40	(-7.711)	40	(-3.568)	(0.009)	
Expenditure female labor	1067	6.196	563	9.526	504	2.475	8.226***	7.051***
	80	(-1.379)	40	(-2.371)	40	(-0.704)	(0.005)	
Expenditure male labor	1429	53.858	748	58.622	681	48.625	3.205*	9.998*
main crop	80	(-2.903)	40	(-3.585)	40	(-4.326)	(0.077)	
Expenditure female labor	1429	48.85	748	46.412	681	51.528	0.18	-5.115
main crop	80	(-6.03)	40	(-8.037)	40	(-9.09)	(0.673)	
Irrigation cost	1429	20.133	748	19.765	681	20.536	0.033	-0.771
	80	(-2.163)	40	(-3.326)	40	(-2.722)	(0.857)	
Irrigation cost main crop	1429	17.371	748	16.541	681	18.283	0.274	-1.741
	80	(-1.693)	40	(-2.588)	40	(-2.124)	(0.602)	
Seed cost	1429	42.759	748	36.992	681	49.094	1.148	-12.101
	80	(-5.666)	40	(-6.951)	40	(-8.985)	(0.287)	
Seed cost main crop	1429	40.888	748	35.181	681	47.157	1.264	-11.976
	80	(-5.345)	40	(-6.442)	40	(-8.562)	(0.264)	

(Contd.)

		(1) All		(2) Loan		(3) Control	F-test for balance across all groups	(2)-(3) Pairwise t-test
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	F-stat/P-value	Mean difference
Number crops	1429	2.373	748	2.412	681	2.33	1.519	0.081
	80	(-0.034)	40	(-0.051)	40	(-0.042)	(0.221)	
Fertilizer cost	1429	101.535	748	92.54	681	111.415	0.482	-18.875
	80	(-13.58)	40	(-17.569)	40	(-20.965)	(0.49)	
Fertilizer cost main crop	1429	80.353	748	65.773	681	96.368	2.757	-30.595
	80	(-9.085)	40	(-7.938)	40	(-16.752)	(0.101)	
Rental cost machinery	1429	55.082	748	54.524	681	55.695	0.01	-1.171
	80	(-5.94)	40	(-8.3)	40	(-8.613)	(0.922)	
Rental cost machinery main crop	1429	47.103	748	43.396	681	51.175	0.948	-7.78
	80	(-3.896)	40	(-3.725)	40	(-7.124)	(0.333)	
Land rent	1429	6.076	748	7.41	681	4.611	1.281	2.799
	80	(-1.265)	40	(-1.886)	40	(-1.621)	(0.261)	
Land rent main crop	1429	6.076	748	7.41	681	4.611	1.281	2.799
	80	(-1.265)	40	(-1.886)	40	(-1.621)	(0.261)	
Days own labor (num)	1429	27.769	748	28.263	681	27.226	0.059	1.037
	80	(-2.124)	40	(-2.816)	40	(-3.232)	(0.808)	
Days own labor main crop	1429	25.731	748	26.476	681	24.913	0.172	1.563
	80	(-1.899)	40	(-2.736)	40	(-2.616)	(0.679)	
Days male HH member's labor (num)	1429	17.624	748	16.311	681	19.066	1.231	-2.755
	80	(-1.256)	40	(-1.666)	40	(-1.86)	(0.271)	
Days male HH member's	1429	16.283	748	15.131	681	17.548	1.244	-2.417
labor main crop	80	(-1.1)	40	(-1.535)	40	(-1.548)	(0.268)	
Days female HH member's labor (num)	1429	14.877	748	14.211	681	15.608	0.204	-1.397
	80	(-1.549)	40	(-2.149)	40	(-2.251)	(0.653)	
Days female HH member's	1429	14.076	748	13.357	681	14.865	0.272	-1.508
labor main crop	80	(-1.446)	40	(-1.965)	40	(-2.147)	(0.604)	
Male wage labor (mean) kharif	1287	2.781	675	2.762	612	2.801	0.056	-0.039
	69	(-0.083)	35	(-0.119)	34	(-0.117)	(0.813)	
Female wage labor (mean) kharif	1287	1.986	675	1.999	612	1.972	0.03	0.028
	69	(-0.081)	35	(-0.127)	34	(-0.099)	(0.863)	

Notes: The table reports outcomes measured at baseline for the main crop cultivated during summer of 2022. All expenditure variables are in INR and deflated to real values. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.5: ITT: Loan intervention and agricultural expansion

	Total area cultivated (1)	Real income from farming (2)	Harvest (3)	Yield (Harvest/ area) (4)	Number crops (5)
			Panel A: All crops		
Treatment: Loan	0.400**	679.596	21.208	1.912	0.089**
	(0.188)	(678.788)	(16.477)	(3.287)	(0.042)
Sharpened q-values	0.109	0.316	0.254	0.508	0.109
Observations	1429	1265	1429	1361	1429
Mean dependent variable (Baseline)	2.966	1071.018	35.424	14.394	2.373
Covariates included	Yes	Yes	Yes	Yes	Yes
Block FE	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes
		Panel B: Main crops (paddy/ cotton/ soybean)			
Treatment: Loan	0.125	-93.875	0.253	-1.270	
	(0.188)	(114.717)	(2.464)	(1.852)	
Sharpened q-values	1	1	1	1	
Observations	1216	847	802	802	
Mean dependent variable (Baseline)	3.049	1202.374	30.595	10.010	
Covariates included	Yes	Yes	Yes	Yes	
Block FE	Yes	Yes	Yes	Yes	
Survey month FE	Yes	Yes	Yes	Yes	

Notes: The table reports the ITT effect of the loan intervention on agricultural expansion, output and productivity during the summer agricultural season. Panel A includes results for all crops and panel B incudes only for the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.6: ITT: Loan intervention and expenditure on inputs

	Total expenditure (real terms)				
	Irrigation	Seeds	Fertilizer	Machines	Rent
	(1)	(2)	(3)	(4)	(5)
Panel A: All crops					
Treatment: Loan	5.326	5.898	33.777	-7.962	3.480
	(3.383)	(5.241)	(35.955)	(11.915)	(6.724)
Sharpened q-values	1	1	1	1	1
Observations	1429	1429	1429	1429	1429
Mean dependent variable (Baseline)	20.133	42.759	101.535	55.082	6.076
Covariates included	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes
Panel B: Main crops (paddy/ cotton/ soybean)					
Treatment: Loan	3.443	4.064	-9.244	-16.153	3.458
	(3.384)	(4.735)	(14.561)	(12.033)	(6.469)
Sharpened q-values	1	1	1	1	1
Observations	1429	1429	1429	1429	1429
Mean dependent variable (Baseline)	17.371	40.888	80.353	47.103	6.076
Covariates included	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on purchase of inputs for cultivation of the main crop during during the summer agricultural season. Panel A includes results for all crops and panel B incudes only for the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.7: ITT: Loan intervention and panel respondent's report of labor utilization

	All crops			Main crops		
	Own labor	Male household member	Female household member	Own labor	Male household member	Female household member
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Loan	-0.448	0.426	0.901	-3.920	-2.110	0.188
	(4.652)	(3.419)	(3.107)	(4.625)	(3.087)	(3.000)
Sharpened q-values	1	1	1	1	1	1
Observations	1429	1429	1429	1429	1429	1429
Mean dependent variable (Baseline)	27.769	17.624	14.877	25.731	16.283	14.076
Covariates included	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on number of days of labor utilization during the summer agricultural season. Columns 1-3 estimate for all crops and columns 4-6 are restricted to the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.8: ITT: Loan intervention and wages

	Aggregate real wages		Total real expenditure (main crops)	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treatment: Loan	-0.034	0.004	-12.561	-6.715
	(0.065)	(0.076)	(15.768)	(12.137)
Sharpened q-values	1	1	1	1
Observations	1161	1160	1429	1429
Mean dependent variable (Baseline)	2.781	1.986	53.858	48.850
Covariates included	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on wages and spending on labor during the summer agricultural season. Columns 1 and 2 report results of all crops and columns 3 and 4 include only the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.9: LATE: Use of formal loans

	Seeds	Pesticides	Labor	Rent machinery	Repayment of another loan
	(1)	(2)	(3)	(4)	(5)
Received loan product	-0.041	-0.196	-0.016	-0.193	-0.318**
	(0.112)	(0.124)	(0.184)	(0.152)	(0.155)
Constant	0.465**	1.419***	1.025***	1.440***	0.175
	(0.235)	(0.354)	(0.326)	(0.298)	(0.299)
Sharpened q value	0.556	0.296	0.596	0.377	0.258
Observations	608	608	608	608	608
Control group mean	0.896	0.906	0.625	0.802	0.432
Covariates included	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of use of formal loans on categories listed as column titles. Each dependent variable is binary. The information was extracted from the endline survey where households were asked how they spent their formal loan(s). The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The model includes covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.10: LATE: Transactions from individual bank account

	Withdraw weekly (1)	Deposit weekly (2)	Receive money weekly (3)
Panel A: Women			
Received loan	0.004	0.037**	0.004
product (γ_F)	(0.010)	(0.018)	(0.013)
Constant	0.089**	0.074	0.005
	(0.044)	(0.067)	(0.032)
Sharpened q-values	0.966	0.122	0.966
Observations	968	968	968
Control group mean	0.006	0.008	0.008
Panel B: Men			
Received loan	-0.105**	0.020	-0.030
product (γ_M)	(0.049)	(0.030)	(0.026)
Constant	0.037	-0.013	0.068
	(0.055)	(0.037)	(0.042)
Sharpened q-values	0.107	0.495	0.332
Observations	1207	1207	1207
Control group mean	0.067	0.045	0.031
Difference ($\gamma_F - \gamma_M$)	0.110	0.017	0.034
p-value of test ($\gamma_F = \gamma_M$)	0.029	0.615	0.239
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on household members' weekly deposit, withdrawal and money receipt in their individual bank account. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Week is the highest frequency of account activity measured in the survey. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.11: LATE: Any time allocation (peak agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan	-0.373***	-0.098	-0.240**	0.064	0.062	0.196
product (γ_F)	(0.122)	(0.074)	(0.107)	(0.118)	(0.052)	(0.144)
Constant	0.347*	0.188*	-0.194	-0.139	1.033***	0.750***
	(0.204)	(0.108)	(0.181)	(0.198)	(0.046)	(0.170)
Sharpened q-values	0.013	0.234	0.067	0.396	0.234	0.234
Observations	1119	1119	1119	1119	453	443
Control group mean	0.619	0.174	0.261	0.351	0.975	0.911
Panel B: Men						
Received loan	-0.102	-0.101	-0.016	0.012	0.226*	0.166**
product (γ_M)	(0.103)	(0.077)	(0.086)	(0.106)	(0.119)	(0.076)
Constant	0.680***	-0.064	0.289	0.709***	1.242***	1.007***
	(0.193)	(0.105)	(0.198)	(0.171)	(0.152)	(0.142)
Sharpened q-values	1	0.924	1	1	1	0.22
Observations	1353	1353	1353	1353	311	528
Control group mean	0.756	0.199	0.232	0.344	0.904	0.922
Difference ($\gamma_F - \gamma_M$)	-0.272	0.003	-0.225	0.052	-0.164	0.030
p-value of test ($\gamma_F = \gamma_M$)	0.088	0.979	0.102	0.744	0.207	0.855
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan product on any time allocation to activities (listed as column titles) in a busy day during the agricultural season. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 5 includes cooking, cleaning, fetching firewood or water, and caring for children or others. Column 6 include eating, sleeping, and leisure such as visiting neighbors, watching TV etc. The sample is restricted to women in the endline survey in Panel A and men in Panel B. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.12: LATE: Any time allocation (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan	-0.300**	-0.091	-0.077	0.057	0.019	0.128**
product (γ_F)	(0.120)	(0.082)	(0.108)	(0.129)	(0.026)	(0.065)
Constant	0.307	0.087	-0.317*	-0.154	0.998***	1.001***
	(0.199)	(0.121)	(0.182)	(0.224)	(0.020)	(0.082)
Sharpened q-values	0.078	0.55	0.611	0.782	0.611	0.137
Observations	1081	1081	1081	1081	416	408
Control group mean	0.563	0.183	0.222	0.313	0.990	0.947
Panel B: Men						
Received loan	-0.049	-0.129	0.039	0.031	0.039	0.101*
product (γ_M)	(0.098)	(0.092)	(0.116)	(0.105)	(0.091)	(0.059)
Constant	0.761***	-0.063	0.175	0.721***	1.319***	0.932***
	(0.192)	(0.124)	(0.197)	(0.173)	(0.099)	(0.115)
Sharpened q-values	1	0.902	1	1	1	0.902
Observations	1345	1345	1345	1345	321	536
Control group mean	0.708	0.215	0.265	0.329	0.949	0.951
Difference ($\gamma_F - \gamma_M$)	-0.251	0.038	-0.116	0.026	-0.019	0.026
p-value of test ($\gamma_F = \gamma_M$)	0.104	0.755	0.463	0.874	0.839	0.765
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan product on any time allocation to activities (listed as column titles) in the day prior to survey. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 5 includes cooking, cleaning, fetching firewood or water, and caring for children or others. Column 6 include eating, sleeping, and leisure such as visiting neighbors, watching TV etc. The sample is restricted to women in the endline survey in Panel A and men in Panel B. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table A.13: LATE: Loan intervention and joint decision making with spouse

	Farming	Livestock & Poultry	Non agriculture	Household purchases
	(1)	(2)	(3)	(4)
Panel A: Women				
Received loan product	-0.081	0.139*	-0.151	0.203*
	(0.127)	(0.073)	(0.104)	(0.110)
Constant	-0.628***	-0.133	-0.411**	-0.500***
	(0.225)	(0.116)	(0.203)	(0.193)
Sharpened q-values	0.248	0.153	0.153	0.153
Observations	1126	1126	1126	1126
Control group mean	0.350	0.088	0.233	0.295
Panel B: Men				
Received loan product	0.042	0.095	0.021	-0.027
	(0.105)	(0.074)	(0.094)	(0.096)
Constant	-0.375***	-0.288***	-0.149*	-0.003
	(0.136)	(0.109)	(0.090)	(0.115)
Sharpened q-values	1	1	1	1
Observations	1364	1364	1364	1364
Control group mean	0.350	0.088	0.233	0.295
Difference ($\gamma_F - \gamma_M$)	-0.124	0.045	-0.172	0.230
p-value of test ($\gamma_F = \gamma_M$)	0.452	0.669	0.221	0.116
Baseline Controls				
Block fixed effects	Yes	Yes	Yes	Yes
Panel C: Concordance between spouses on joint decision making				
Received loan product	-0.134	-0.043	-0.007	0.061
	(0.092)	(0.061)	(0.044)	(0.054)
Constant	0.088	-0.100	-0.063	0.167
	(0.132)	(0.072)	(0.059)	(0.118)
Sharpened q-values	1	1	1	1
Observations	499	499	499	499
Control group mean	0.106	0.048	0.031	0.035
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on decisions made jointly with their spouse. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The outcome variables are binary and take the value 0 if the woman is not involved in the decision or makes the decision on her own. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 4 includes decisions on purchase of large items such as bicycles, land and transport vehicles as well as routine items such as food. The sample is restricted to women in the endline survey in Panel A and men in Panel B. Therefore, only one observation per household is analyzed. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.14: LATE results by state: Loan product and women's individual bank account and debit card use

	Own bank account (1)	Use debit card alone (2)	Share debit card (3)
Panel A: Odisha			
Received loan product	0.115	0.898***	-0.250**
	(0.087)	(0.160)	(0.117)
Constant	0.598***	0.701***	0.046
	(0.157)	(0.264)	(0.179)
Sharpened q-values	0.066	0.001	0.035
Observations	736	657	657
Control group mean	0.866	0.193	0.408
Panel B: Maharashtra			
Received loan product	-0.233	0.700***	0.068
	(0.207)	(0.167)	(0.108)
Constant	0.986***	0.922***	0.306
	(0.358)	(0.289)	(0.206)
Sharpened q-values	0.354	0.001	0.544
Observations	390	311	311
Control group mean	0.842	0.167	0.167

Notes: The table reports LATE estimates of treatment on women's bank account ownership and use of debit card. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Columns 2 and 3 are conditional on respondent owning an individual bank account. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.15: LATE results by state: Loan product and men's individual bank account and debit card use

	Own bank account	Use debit card alone	Share debit card
	(1)	(2)	(3)
	Panel A: Odisha		
Received loan product	0.093	0.246*	0.055
	(0.073)	(0.141)	(0.143)
Constant	0.991***	0.684***	0.098
	(0.079)	(0.259)	(0.187)
Sharpened q-values	0.322	0.322	0.429
Observations	726	697	697
Control group mean	0.941	0.496	0.331
	(1)	(2)	(3)
	Panel B: Maharashtra		
Received loan product	0.187	0.300**	0.013
	(0.116)	(0.117)	(0.116)
Constant	0.462**	1.581***	-0.253
	(0.217)	(0.231)	(0.178)
Sharpened q-values	0.12	0.031	0.438
Observations	638	510	510
Control group mean	0.842	0.167	0.167

Notes: The table reports LATE estimates of treatment on men's bank account ownership and use of debit card. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Columns 2 and 3 are conditional on respondent owning an individual bank account. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.16: LATE results by state: Loan product, women's mobile phone ownership and digital financial services

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
Panel A: Odisha				
Received loan product (γ_F)	0.154	0.392**	1.158***	0.962***
	(0.198)	(0.154)	(0.163)	(0.132)
Sharpened q-values	0.123	0.008	0.001	0.001
Observations	377	423	657	657
Control group mean	0.397	0.315	0.232	0.193
Panel B: Maharashtra				
Received loan product (γ_F)	0.229	-0.018	0.803***	0.876***
	(0.151)	(0.148)	(0.146)	(0.146)
Sharpened q-values	0.095	0.293	0.001	0.001
Observations	248	240	311	311
Control group mean	0.228	0.259	0.183	0.140

Notes: The table reports LATE estimates of treatment on women's smartphone ownership and uptake of digital financial services. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. The analysis is conditional on respondent owning an individual bank account. Panel A restricts the sample to Odisha and panel B to Maharashtra. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.17: LATE results by state: Loan product, men's mobile phone ownership and digital financial services

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
		Panel A: Odisha		
Received loan	-0.212	0.129	0.639***	0.496***
product (γ_F)	(0.252)	(0.195)	(0.137)	(0.150)
Sharpened q-values	0.123	0.008	0.001	0.001
Observations	391	453	697	697
Control group mean	0.428	0.581	0.583	0.535
		Panel B: Maharashtra		
Received loan	0.017	0.048	0.338***	0.342***
product (γ_F)	(0.092)	(0.127)	(0.076)	(0.074)
Sharpened q-values	0.095	0.293	0.001	0.001
Observations	341	421	510	510
Control group mean	0.815	0.833	0.638	0.622

Notes: The table reports LATE estimates of treatment on men's smartphone ownership and uptake of digital financial services. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. The analysis is conditional on respondent owning an individual bank account. Panel A restricts the sample to Odisha and panel B to Maharashtra. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.18: LATE results by state: Loan product and women's time use (peak time in agricultural season)

	Farming (1)	Livestock & Poultry (2)	Non-agriculture (3)	Household purchases (4)	Domestic work (5)	Personal activity (6)
Panel A: Odisha						
Received loan product	0.498	-0.039	0.220	0.648*	2.031	1.944**
	(0.971)	(0.228)	(0.223)	(0.336)	(1.378)	(0.760)
Constant	7.479***	0.122	-0.045	0.503	5.183***	-1.392**
	(1.016)	(0.390)	(0.551)	(0.373)	(1.321)	(0.704)
Sharpened q-values	0.575	0.763	0.393	0.157	0.232	0.071
Observations	734	734	734	734	190	195
Control group mean	2.724	0.380	0.214	0.298	2.601	0.908
Panel B: Maharashtra						
Received loan product	-4.631***	-0.480**	-4.796***	-0.794***	-0.670*	0.397*
	(1.283)	(0.240)	(0.798)	(0.214)	(0.367)	(0.236)
Constant	1.391	0.598*	-2.588*	-2.155***	3.630***	2.599***
	(1.488)	(0.343)	(1.338)	(0.436)	(0.757)	(0.561)
Sharpened q-values	0.001	0.036	0.001	0.001	0.043	0.049
Observations	385	385	385	385	263	248
Control group mean	3.806	0.315	3.167	1.306	3.429	1.988
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on hours spent by women in different activities (listed as column titles) in a busy day during the agricultural season. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables are listed as column titles. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 5 includes cooking, cleaning, fetching firewood or water, and caring for children or others. Column 6 include eating, sleeping, and leisure such as visiting neighbors, watching TV etc. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.19: LATE results by state: Loan product and men's time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Odisha						
Received loan product	0.798	0.183	0.808***	0.050	3.394*	2.155***
	(1.320)	(0.237)	(0.256)	(0.269)	(1.945)	(0.798)
Constant	3.394**	-0.457	0.539**	0.766*	6.155***	-0.216
	(1.517)	(0.361)	(0.263)	(0.402)	(2.206)	(1.233)
Sharpened q-values	0.488	0.488	0.013	0.744	0.122	0.018
Observations	723	723	723	723	118	186
Control group mean	3.757	0.405	0.171	0.409	1.147	0.804
Panel B: Maharashtra						
Received loan product	0.124	-0.759**	-1.581*	-0.776***	-0.176	-0.020
	(0.795)	(0.320)	(0.842)	(0.228)	(0.535)	(0.167)
Constant	2.138	-0.181	2.507	1.550**	1.471*	1.918***
	(1.656)	(0.552)	(1.825)	(0.722)	(0.857)	(0.400)
Sharpened q-values	0.824	0.048	0.087	0.007	0.824	0.824
Observations	630	630	630	630	193	342
Control group mean	4.461	0.558	2.124	0.895	1.740	1.788
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on hours spent by men in different activities (listed as column titles) in a busy day during the agricultural season. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables are listed as column titles. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 5 includes cooking, cleaning, fetching firewood or water, and caring for children or others. Column 6 include eating, sleeping, and leisure such as visiting neighbors, watching TV etc. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.20: LATE results by state: Loan product and women's time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Odisha						
Received loan product	0.126	-0.094	1.039***	0.469	0.053	1.868***
	(0.759)	(0.238)	(0.324)	(0.286)	(1.093)	(0.630)
Constant	4.264***	-0.066	-0.881	0.099	5.838***	0.913
	(0.897)	(0.407)	(0.555)	(0.332)	(1.034)	(0.715)
Sharpened q-values	0.925	0.925	0.007	0.156	0.925	0.008
Observations	735	735	735	735	191	199
Control group mean	1.955	0.416	0.334	0.269	3.290	1.235
Panel B: Maharashtra						
Received loan product	-4.054***	-0.532	-3.641***	-0.939	-0.711**	0.590*
	(0.915)	(0.354)	(0.623)	(0.885)	(0.311)	(0.311)
Constant	0.819	0.464	-1.870	-7.442***	4.166***	1.858***
	(1.270)	(0.370)	(1.286)	(1.326)	(0.841)	(0.554)
Sharpened q-values	0.001	0.088	0.001	0.107	0.031	0.047
Observations	346	346	346	346	225	209
Control group mean	2.831	0.373	2.209	3.469	3.654	2.099
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on hours spent by women in different activities (listed as column titles) in the day prior to survey. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables are listed as column titles. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 5 includes cooking, cleaning, fetching firewood or water, and caring for children or others. Column 6 include eating, sleeping, and leisure such as visiting neighbors, watching TV etc. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.21: LATE results by state: Loan product and men's time use (yesterday)

	Farming (1)	Livestock & Poultry (2)	Non-agriculture (3)	Household purchases (4)	Domestic work (5)	Personal activity (6)
Panel A: Odisha						
Received loan product	0.633	0.008	0.831	0.155	1.155	1.944**
	(1.015)	(0.311)	(0.763)	(0.232)	(1.139)	(0.805)
Constant	3.317***	0.015	1.603**	1.259**	5.853***	1.084
	(1.272)	(0.541)	(0.658)	(0.539)	(1.859)	(1.153)
Sharpened q-values	1	1	1	1	1	0.107
Observations	718	718	718	718	127	193
Control group mean	2.424	0.478	0.874	0.375	1.694	1.042
Panel B: Maharashtra						
Received loan product	0.051	-0.692**	-1.006	-0.548	-0.687	-0.414
	(0.772)	(0.338)	(0.676)	(0.884)	(0.495)	(0.309)
Constant	1.996	-0.409	1.402	3.915**	2.204***	3.219***
	(1.348)	(0.551)	(1.505)	(1.885)	(0.795)	(0.594)
Sharpened q-values	0.462	0.327	0.327	0.37	0.327	0.327
Observations	627	627	627	627	194	343
Control group mean	3.187	0.557	1.599	2.322	1.917	2.255
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on hours spent by men in different activities (listed as column titles) in the day prior to survey. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables are listed as column titles. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 5 includes cooking, cleaning, fetching firewood or water, and caring for children or others. Column 6 include eating, sleeping, and leisure such as visiting neighbors, watching TV etc. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.22: LATE results by state: Loan product and women's involvement in decision-making

	Share of decisions made (1)	Farming (2)	Livestock & Poultry (3)	Non agriculture (4)	Household purchases (5)	Private consumption (6)	Domestic work (7)	Personal activities (8)
Panel A: Odisha								
Received loan product	0.319***	0.268**	0.190	0.329***	0.468***	0.438***	0.489***	0.036
	(0.077)	(0.113)	(0.137)	(0.114)	(0.161)	(0.139)	(0.159)	(0.125)
Constant	0.278*	1.314***	0.102	-0.143	0.409*	0.499	0.537	1.149***
	(0.156)	(0.160)	(0.229)	(0.227)	(0.248)	(0.380)	(0.444)	(0.199)
Sharpened q-values	0.001	0.01	0.051	0.006	0.006	0.005	0.005	0.238
Observations	736	736	736	736	736	736	736	736
Control group mean	0.248	0.671	0.267	0.148	0.318	0.284	0.290	0.635
Panel B: Maharashtra								
Received loan product	-0.342***	-0.894***	-0.156**	-0.836***	-0.178	-0.221	-0.200	-0.276*
	(0.058)	(0.190)	(0.077)	(0.130)	(0.126)	(0.143)	(0.146)	(0.149)
Constant	-0.181**	0.197	0.177*	-0.325	-0.764***	-0.637***	-0.544***	0.377*
	(0.090)	(0.256)	(0.099)	(0.222)	(0.243)	(0.199)	(0.176)	(0.215)
Sharpened q-values	0.001	0.001	0.059	0.001	0.097	0.097	0.097	0.068
Observations	390	390	390	390	390	390	390	390
Control group mean	0.429	0.701	0.109	0.552	0.688	0.792	0.751	0.900
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on involvement in decision making (listed as column titles). The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. Column 1 is the share of total decisions. Columns 2-8 are binary variables of production and non-production activities described in Table 2. The sample is restricted to women in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.23: LATE results by state: Loan product and men's involvement in decision-making

	Share of decisions made (1)	Farming (2)	Livestock & Poultry (3)	Non agriculture (4)	Household purchases (5)	Private consumption (6)	Domestic work (7)	Personal activities (8)
Panel A: Odisha								
Received loan product	0.210*	-0.185***	0.113	0.304	0.061	0.402**	0.493***	0.303***
	(0.111)	(0.061)	(0.184)	(0.208)	(0.209)	(0.165)	(0.185)	(0.117)
Constant	0.203**	0.673***	0.052	0.093	0.571**	0.197	0.422***	0.013
	(0.103)	(0.204)	(0.228)	(0.160)	(0.257)	(0.124)	(0.125)	(0.130)
Sharpened q-values	0.48	0.017	0.236	0.101	0.24	0.027	0.024	0.024
Observations	726	726	726	726	726	726	726	726
Control group mean	0.284	0.956	0.363	0.263	0.459	0.159	0.215	0.107
Panel B: Maharashtra								
Received loan product	-0.112***	0.009	-0.285***	-0.280**	-0.047	-0.259*	-0.167	0.033
	(0.042)	(0.119)	(0.086)	(0.124)	(0.135)	(0.135)	(0.134)	(0.129)
Constant	0.234**	0.496**	0.045	0.432	0.842***	0.339	0.625***	0.330
	(0.094)	(0.249)	(0.192)	(0.312)	(0.243)	(0.234)	(0.224)	(0.224)
Sharpened q-values	0.029	0.543	0.009	0.051	0.52	0.074	0.206	0.52
Observations	638	638	638	638	638	638	638	638
Control group mean	0.264	0.746	0.219	0.370	0.491	0.399	0.609	0.299
Baseline Controls								
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on involvement in decision making (listed as column titles). The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. Column 1 is the share of total decisions. Columns 2-8 are binary variables of production and non-production activities described in Table 2. The sample is restricted to men in the endline survey in Odisha (Panel A) and Maharashtra (Panel B). The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in brackets. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.24: LATE: Heterogeneous effects of ex-ante asset ownership on bank account and debit card use

	Own bank account (1)	Use debit card alone (2)	Share debit card (3)
	Panel A: Women		
Received loan * Asset quintile	0.037	0.017	-0.064
	(0.050)	(0.063)	(0.057)
Received Loan	-0.139	0.756***	0.098
	(0.217)	(0.279)	(0.244)
Asset quintile	-0.020	-0.028	0.022
	(0.018)	(0.018)	(0.022)
Constant	0.483**	0.787***	-0.015
	(0.236)	(0.216)	(0.218)
Sharpened q-values	1	1	1
Observations	1126	968	968
Control group mean	0.857	0.178	0.309
	Panel B: Men		
Received loan * Asset quintile	0.041	-0.101*	0.068
	(0.045)	(0.056)	(0.054)
Received Loan	0.020	0.607***	-0.183
	(0.165)	(0.192)	(0.194)
Asset quintile	0.002	0.018	-0.012
	(0.015)	(0.019)	(0.019)
Constant	0.701***	1.093***	0.003
	(0.177)	(0.184)	(0.156)
Sharpened q-values	1	1	1
Observations	1364	1207	1207
Control group mean	0.836	0.520	0.227
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the asset quintile of the household from baseline to instrument for heterogeneous effect of asset ownership in loan receiving households. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.25: LATE: Heterogeneous effects of ex-ante asset ownership on mobile phone and digital finance

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
		Panel A: Women		
Received loan *	-0.038	-0.002	0.001	0.005
Asset quintile	(0.071)	(0.067)	(0.060)	(0.058)
Received Loan	0.374	0.251	1.028***	0.923***
	(0.280)	(0.272)	(0.258)	(0.235)
Asset quintile	-0.036	-0.008	-0.028*	-0.020
	(0.023)	(0.024)	(0.017)	(0.015)
Constant	0.803***	0.927***	0.498**	0.634***
	(0.264)	(0.267)	(0.227)	(0.203)
Sharpened q-values	1	1	1	1
Observations	625	663	968	968
Control group mean	0.337	0.300	0.219	0.184
		Panel B: Men		
Received loan *	0.050	-0.048	-0.011	0.026
Asset quintile	(0.061)	(0.068)	(0.054)	(0.052)
Received Loan	-0.240	0.185	0.527***	0.363**
	(0.196)	(0.233)	(0.169)	(0.165)
Asset quintile	-0.038	-0.014	-0.005	-0.013
	(0.024)	(0.026)	(0.021)	(0.019)
Constant	0.270	0.899***	1.296***	1.419***
	(0.190)	(0.220)	(0.167)	(0.149)
Sharpened q-values	1	1	1	1
Observations	732	874	1207	1207
Control group mean	0.647	0.717	0.609	0.582
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the asset quintile of the household from baseline to instrument for heterogeneous effect of asset ownership in loan receiving households. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.26: LATE: Heterogeneous effects of ex-ante asset ownership on time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	0.043	-0.058	0.070	-0.081	-0.490**	0.016
Asset quintile	(0.418)	(0.072)	(0.329)	(0.141)	(0.227)	(0.158)
Received Loan	-1.722	-0.003	-2.119*	0.223	1.487	0.848
	(1.790)	(0.292)	(1.125)	(0.573)	(0.931)	(0.654)
Asset quintile	-0.054	0.023	0.053	0.013	0.033	0.055
	(0.121)	(0.038)	(0.120)	(0.049)	(0.069)	(0.047)
Constant	1.669	0.122	-1.587	-1.145***	4.391***	2.048***
	(1.161)	(0.316)	(0.990)	(0.423)	(0.758)	(0.590)
Sharpened q-values	1	1	1	1	0.229	1
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan *	-0.405	-0.247*	0.252	-0.354***	0.385	0.228*
Asset quintile	(0.453)	(0.148)	(0.349)	(0.130)	(0.339)	(0.129)
Received Loan	1.746	0.366	-1.268	0.594*	-0.229	-0.230
	(1.369)	(0.389)	(1.359)	(0.318)	(1.061)	(0.332)
Asset quintile	-0.044	0.060	-0.091	0.024	-0.058	-0.127***
	(0.166)	(0.067)	(0.106)	(0.046)	(0.101)	(0.047)
Constant	2.668**	-0.725	2.012	0.326	2.415**	1.801***
	(1.227)	(0.513)	(1.264)	(0.389)	(0.955)	(0.444)
Sharpened q-values	0.287	0.191	0.309	0.038	0.238	0.191
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the asset quintile of the household from baseline to instrument for heterogeneous effect of asset ownership in loan receiving households. Dependent variables are number of hours spent in activity and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ $p< 0.1$, $**$ $p< .05$, $***$ $p< .01$

Table A.27: LATE: Heterogeneous effects of ex-ante asset ownership on time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
			Panel A: Women			
Received loan *	0.434	-0.088	0.304	-0.248	-0.104	-0.009
Asset quintile	(0.357)	(0.088)	(0.279)	(0.343)	(0.252)	(0.186)
Received Loan	-2.826**	0.020	-1.702*	0.502	-0.147	1.033
	(1.371)	(0.335)	(0.886)	(1.319)	(0.997)	(0.721)
Asset quintile	-0.174	0.023	-0.059	0.070	0.054	0.075
	(0.106)	(0.045)	(0.094)	(0.099)	(0.073)	(0.054)
Constant	0.910	-0.169	-1.483*	-1.413	5.409***	2.088***
	(0.974)	(0.295)	(0.831)	(1.180)	(0.815)	(0.407)
Sharpened q-values	1	1	1	1	1	1
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
		Panel B: Men				
Received loan *	-0.323	-0.317*	-0.022	-0.629*	0.441	0.445**
Asset quintile	(0.387)	(0.163)	(0.294)	(0.361)	(0.317)	(0.208)
Received Loan	1.283	0.516	-0.210	1.538	-1.347	-1.170*
	(1.182)	(0.397)	(1.099)	(1.178)	(0.955)	(0.667)
Asset quintile	0.009	0.094	-0.061	-0.064	-0.090	-0.189**
	(0.129)	(0.078)	(0.090)	(0.091)	(0.108)	(0.076)
Constant	1.840*	-0.689	1.168	3.413**	3.641***	3.122***
	(1.033)	(0.533)	(1.109)	(1.437)	(0.952)	(0.592)
Sharpened q-values	0.319	0.185	0.457	0.185	0.197	0.185
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the asset quintile of the household from baseline to instrument for heterogeneous effect of asset ownership in loan receiving households. Dependent variables are number of hours spent in activity and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.28: LATE: Heterogeneous effects of ex-ante asset ownership on involvement in decision-making

	Share of decisions made	Farming	Livestock & Poultry	Non agriculture	Household purchases	Private consumption	Domestic work	Personal activities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Women								
Received loan *	0.049	0.078	0.036	0.069	0.029	0.014	0.068	0.076
Asset quintile	(0.039)	(0.071)	(0.032)	(0.063)	(0.068)	(0.070)	(0.070)	(0.056)
Received Loan	-0.108	-0.463*	-0.049	-0.365*	0.079	0.108	-0.023	-0.346
	(0.126)	(0.258)	(0.137)	(0.213)	(0.258)	(0.286)	(0.272)	(0.219)
Asset quintile	-0.012	-0.031	-0.008	-0.008	-0.011	-0.012	-0.033*	-0.028
	(0.012)	(0.023)	(0.015)	(0.021)	(0.021)	(0.022)	(0.020)	(0.019)
Constant	-0.024	0.206	0.010	-0.311	-0.041	0.088	0.172	1.079***
	(0.096)	(0.201)	(0.121)	(0.211)	(0.199)	(0.189)	(0.211)	(0.175)
Sharpened q-values	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan *	-0.023	-0.067	-0.020	-0.003	-0.149**	-0.073	-0.085	-0.049
Asset quintile	(0.024)	(0.051)	(0.055)	(0.065)	(0.061)	(0.060)	(0.065)	(0.046)
Received Loan	0.090	0.142	-0.055	-0.024	0.442**	0.214	0.338	0.281
	(0.072)	(0.137)	(0.154)	(0.210)	(0.203)	(0.227)	(0.215)	(0.182)
Asset quintile	-0.008	-0.018	-0.003	-0.013	-0.018	0.006	-0.003	-0.001
	(0.008)	(0.018)	(0.020)	(0.022)	(0.018)	(0.017)	(0.019)	(0.014)
Constant	0.106	0.539***	-0.092	0.214	0.588***	0.178	0.552***	0.068
	(0.081)	(0.161)	(0.156)	(0.227)	(0.192)	(0.188)	(0.160)	(0.208)
Sharpened q-values	0.658	0.636	0.799	0.83	0.137	0.636	0.636	0.636
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables are listed as column titles. The exogenous variable treatment assignment is interacted with the asset quintile of the household from baseline to instrument for heterogeneous effect of asset ownership in loan receiving households. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.29: LATE: Heterogeneous effects of ex-ante spousal coordination on bank account and debit card use

	Own bank account (1)	Use debit card alone (2)	Share debit card (3)
Panel A: Women			
Received loan * BL	-0.570**	-0.147	0.145
share of joint decisions	(0.244)	(0.609)	(0.440)
Received Loan	0.023	0.834***	-0.121
	(0.103)	(0.124)	(0.104)
Baseline Share	0.102	-0.159	-0.017
of joint decisions	(0.090)	(0.106)	(0.111)
Constant	0.481**	0.722***	0.068
	(0.214)	(0.177)	(0.205)
Sharpened q-values	0.064	1	1
Observations	1126	968	968
Control group mean	0.857	0.178	0.309
Panel B: Men			
Received loan * BL	0.204	0.044	-0.482
share of joint decisions	(0.320)	(0.383)	(0.455)
Received Loan	0.140*	0.317***	0.040
	(0.082)	(0.101)	(0.102)
Baseline Share	-0.280*	-0.187	0.289
of joint decisions	(0.155)	(0.201)	(0.243)
Constant	0.666***	1.239***	-0.122
	(0.139)	(0.178)	(0.133)
Sharpened q-values	1	1	1
Observations	1364	1207	1207
Control group mean	0.836	0.520	0.227
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the share of decisions farmer reported making jointly with spouse in the baseline survey to instrument for heterogeneous effect of ex-ante spousal coordination in loan-receiving households on outcome variables. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. ∗ p< 0.1, ∗∗ p< .05, ∗∗∗ p< .01

Table A.30: LATE: Heterogeneous effects of ex-ante spousal coordination on mobile phone and digital finance

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
Panel A: Women				
Received loan * BL	0.976**	-0.017	-0.022	-0.400
share of joint decisions	(0.421)	(0.377)	(0.493)	(0.379)
Received Loan	0.192	0.244**	1.032***	0.964***
	(0.136)	(0.124)	(0.116)	(0.108)
Baseline Share	-0.249*	-0.057	0.061	0.102
of joint decisions	(0.150)	(0.184)	(0.094)	(0.094)
Constant	0.896***	0.955***	0.476**	0.615***
	(0.245)	(0.245)	(0.208)	(0.192)
Sharpened q-values	0.087	1	1	0.78
Observations	625	663	968	968
Control group mean	0.337	0.300	0.219	0.184
Panel B: Men				
Received loan * BL	-0.437	-0.670	-0.408	-0.531
share of joint decisions	(0.674)	(0.766)	(0.405)	(0.438)
Received Loan	-0.095	0.082	0.525***	0.469***
	(0.126)	(0.117)	(0.080)	(0.086)
Baseline Share	0.509	0.113	0.073	0.211
of joint decisions	(0.360)	(0.404)	(0.218)	(0.244)
Constant	0.168	0.969***	1.300***	1.355***
	(0.163)	(0.173)	(0.161)	(0.145)
Sharpened q-values	1	1	1	1
Observations	732	874	1207	1207
Control group mean	0.647	0.717	0.609	0.582
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the share of decisions farmer reported making jointly with spouse in the baseline survey to instrument for heterogeneous effect of ex-ante spousal coordination in loan-receiving households on outcome variables. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.31: LATE: Heterogeneous effects of ex-ante spousal coordination on time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	3.903	0.515	3.949***	1.935**	7.082***	0.063
Asset quintile	(2.457)	(0.434)	(1.005)	(0.760)	(2.499)	(1.548)
Received Loan	-1.790**	-0.221	-2.216***	-0.184	-0.403	0.903***
	(0.838)	(0.190)	(0.521)	(0.225)	(0.505)	(0.325)
Baseline Share	-1.922*	-0.267	-0.555	-0.461**	-1.565**	-0.143
of joint decisions	(1.115)	(0.217)	(0.383)	(0.224)	(0.690)	(0.526)
Constant	1.469	0.219	-1.633*	-1.056***	4.596***	1.945***
	(1.030)	(0.305)	(0.938)	(0.404)	(0.793)	(0.523)
Sharpened q-values	0.092	0.165	0.001	0.016	0.013	0.393
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan *	-1.905	0.113	3.416**	1.622**	-3.284	1.153
Asset quintile	(2.189)	(0.962)	(1.711)	(0.729)	(3.191)	(1.463)
Received Loan	0.678	-0.384	-0.738	-0.549**	0.997	0.327
	(0.849)	(0.297)	(0.587)	(0.234)	(0.694)	(0.212)
Baseline Share	0.787	0.201	-1.511*	-0.673*	1.804	-0.805
of joint decisions	(1.162)	(0.523)	(0.806)	(0.370)	(1.755)	(0.698)
Constant	2.965***	-0.270	1.952**	0.981**	2.133***	1.639***
	(1.076)	(0.358)	(0.928)	(0.432)	(0.772)	(0.466)
Sharpened q-values	0.527	1	0.161	0.161	0.527	0.527
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the share of decisions farmer reported making jointly with spouse in the baseline survey to instrument for heterogeneous effect of ex-ante spousal coordination in loan-receiving households on outcome variables. Dependent variables are hours spent in activity listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.32: LATE: Heterogeneous effects of ex-ante spousal coordination on time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	1.958	1.098**	2.836***	1.258	4.582**	2.302
Asset quintile	(1.993)	(0.437)	(0.985)	(1.292)	(1.780)	(1.723)
Received Loan	-1.533**	-0.326	-0.957**	-0.401	-0.772	0.939***
	(0.622)	(0.226)	(0.485)	(0.540)	(0.524)	(0.362)
Baseline Share	-1.137	-0.512**	-0.562	-0.215	-0.251	-0.785
of joint decisions	(0.899)	(0.208)	(0.382)	(0.396)	(0.654)	(0.506)
Constant	0.508	-0.119	-1.790**	-1.164	5.172***	2.064***
	(0.856)	(0.279)	(0.783)	(1.188)	(0.802)	(0.347)
Sharpened q-values	0.198	0.025	0.025	0.198	0.025	0.159
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan *	-2.888	0.423	1.978	1.999	-2.738	1.970
Asset quintile	(2.131)	(1.023)	(1.610)	(2.005)	(2.914)	(1.345)
Received Loan	0.489	-0.454	-0.378	-0.430	-0.015	-0.058
	(0.697)	(0.319)	(0.580)	(0.666)	(0.504)	(0.319)
Baseline Share	1.601	0.017	-1.169	-0.888	1.535	-1.371**
of joint decisions	(1.126)	(0.547)	(0.789)	(0.989)	(1.653)	(0.652)
Constant	2.073**	-0.165	1.395	4.537***	3.393***	2.701***
	(0.884)	(0.374)	(0.927)	(1.404)	(0.755)	(0.503)
Sharpened q-values	0.714	0.714	0.714	0.714	0.714	0.714
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the share of decisions farmer reported making jointly with spouse in the baseline survey to instrument for heterogeneous effect of ex-ante spousal coordination in loan-receiving households on outcome variables. Dependent variables are hours spent in activity listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.33: LATE: Heterogeneous effects of ex-ante spousal coordination on involvement in decisions

	Share of decisions made (1)	Farming (2)	Livestock & Poultry (3)	Non agriculture (4)	Household purchases (5)	Private consumption (6)	Domestic work (7)	Personal activities (8)
Panel A: Women								
Received loan * BL	0.629***	0.632	0.404	0.808***	1.052**	0.904**	0.996*	0.611**
share of joint decisions	(0.230)	(0.390)	(0.267)	(0.258)	(0.497)	(0.453)	(0.535)	(0.276)
Received Loan	0.000	-0.260**	0.042	-0.214*	0.087	0.089	0.126	-0.156
	(0.070)	(0.125)	(0.101)	(0.124)	(0.123)	(0.115)	(0.123)	(0.114)
Baseline Share	-0.134**	-0.156	-0.162	-0.057	-0.167	-0.230	-0.267*	-0.027
of joint decisions	(0.066)	(0.165)	(0.123)	(0.106)	(0.181)	(0.146)	(0.145)	(0.126)
Constant	-0.065	0.144	-0.019	-0.366*	-0.079	0.051	0.093	1.001***
	(0.105)	(0.180)	(0.123)	(0.210)	(0.189)	(0.191)	(0.207)	(0.156)
Sharpened q-values	0.015	0.061	0.061	0.011	0.036			
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan * BL	0.144	-0.275	0.118	1.040***	-0.246	0.316	0.242	-0.383
share of joint decisions	(0.169)	(0.273)	(0.408)	(0.372)	(0.434)	(0.418)	(0.406)	(0.360)
Received Loan	0.017	-0.036	-0.130	-0.089	0.021	-0.010	0.081	0.164
	(0.063)	(0.092)	(0.103)	(0.126)	(0.140)	(0.134)	(0.136)	(0.105)
Baseline Share	-0.115	0.102	0.087	-0.540***	0.097	-0.279	-0.235	0.101
of joint decisions	(0.081)	(0.127)	(0.221)	(0.167)	(0.232)	(0.217)	(0.218)	(0.195)
Constant	0.158**	0.610***	-0.049	0.292	0.744***	0.310*	0.681***	0.139
	(0.076)	(0.163)	(0.131)	(0.186)	(0.205)	(0.167)	(0.139)	(0.171)
Sharpened q-values	0.717	0.717	1	0.026	1			
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with the share of decisions farmer reported making jointly with spouse in the baseline survey to instrument for heterogeneous effect of ex-ante spousal coordination in loan-receiving households on outcome variables. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.34: LATE: Heterogeneous effects of caste on individual bank account and debit card use

	Own bank account (1)	Use debit card alone (2)	Share debit card (3)
Panel A: Women			
Received loan * SC/ST	-0.024	0.020	-0.085
	(0.141)	(0.176)	(0.191)
Received Loan	-0.013	0.806***	-0.082
	(0.114)	(0.132)	(0.114)
SC/ST	0.080	-0.059	0.094
	(0.050)	(0.054)	(0.060)
Constant	0.462**	0.735***	0.074
	(0.217)	(0.178)	(0.208)
Sharpened q-values	1	1	1
Observations	1126	968	968
Control group mean	0.857	0.178	0.309
Panel B: Men			
Received loan * SC/ST	-0.051	-0.007	0.193
	(0.127)	(0.172)	(0.153)
Received Loan	0.155**	0.309***	-0.035
	(0.078)	(0.111)	(0.107)
SC/ST	0.006	-0.021	-0.030
	(0.035)	(0.056)	(0.049)
Constant	0.645***	1.236***	-0.112
	(0.152)	(0.181)	(0.140)
Sharpened q-values	0.54	0.507	0.507
Observations	1364	1207	1207
Control group mean	0.836	0.520	0.227
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with binary variable of household belonging to Scheduled Caste/ Scheduled Tribe to instrument for heterogeneous effect of caste in loan receiving households. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.35: LATE: Heterogeneous effects of caste on mobile phone and digital financial services

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
Panel A: Women				
Received loan * SC/ST	-0.356*	0.049	0.099	0.172
	(0.188)	(0.200)	(0.144)	(0.169)
Received Loan	0.359**	0.222*	1.000***	0.881***
	(0.148)	(0.126)	(0.123)	(0.130)
SC/ST	-0.050	-0.045	-0.046	-0.047
	(0.059)	(0.073)	(0.051)	(0.042)
Constant	0.887***	0.952***	0.465**	0.598***
	(0.239)	(0.256)	(0.207)	(0.191)
Sharpened q-values	0.309	1	0.973	0.87
Observations	625	663	968	968
Control group mean	0.337	0.300	0.219	0.184
Panel B: Men				
Received loan * SC/ST	-0.326*	-0.149	0.189	0.228
	(0.193)	(0.149)	(0.154)	(0.169)
Received Loan	-0.011	0.090	0.441***	0.374***
	(0.118)	(0.126)	(0.084)	(0.090)
SC/ST	0.061	0.087	-0.014	-0.021
	(0.057)	(0.054)	(0.049)	(0.052)
Constant	0.211	0.976***	1.305***	1.363***
	(0.164)	(0.171)	(0.172)	(0.159)
Observations	732	874	1207	1207
Control group mean	0.647	0.717	0.609	0.582
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with binary variable of household belonging to Scheduled Caste/ Scheduled Tribe to instrument for heterogeneous effect of caste in loan receiving households. Dependent variables are binary and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.36: LATE: Heterogeneous effects of caste on time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan * SC/ST	1.617	-0.482*	0.627	0.383	-0.164	0.625
	(1.380)	(0.290)	(0.730)	(0.348)	(0.930)	(0.627)
Received Loan	-2.112***	-0.038	-2.105***	-0.169	-0.012	0.670**
	(0.809)	(0.199)	(0.544)	(0.242)	(0.502)	(0.282)
SC/ST	0.130	0.125	0.130	0.068	0.173	-0.148
	(0.400)	(0.100)	(0.291)	(0.124)	(0.253)	(0.122)
Constant	1.557	0.271	-1.524	-1.002**	4.996***	1.824***
	(1.153)	(0.331)	(1.001)	(0.403)	(0.835)	(0.518)
Sharpened q-values	0.884	0.884	0.884	0.884	0.884	0.884
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan * SC/ST	0.310	-0.523	-0.043	0.271	2.290*	0.715
	(1.572)	(0.680)	(0.942)	(0.404)	(1.315)	(0.570)
Received Loan	0.472	-0.217	-0.518	-0.523**	0.185	0.197
	(0.794)	(0.185)	(0.538)	(0.241)	(0.502)	(0.213)
SC/ST	-0.508	0.234	-0.140	-0.177	-0.431*	-0.303**
	(0.521)	(0.292)	(0.325)	(0.129)	(0.261)	(0.139)
Constant	3.069***	-0.237	1.736*	0.866**	1.448*	1.425***
	(1.047)	(0.369)	(0.920)	(0.435)	(0.862)	(0.510)
Sharpened q-values	1	1	1	1	0.946	1
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with binary variable of household belonging to Scheduled Caste/ Scheduled Tribe to instrument for heterogeneous effect of caste in loan receiving households. Dependent variables are number of hours spent in activity and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. ∗ p< 0.1, ∗∗ p< .05, ∗ ∗ ∗ p< .01

Table A.37: LATE: Heterogeneous effects of caste time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	1.782	-0.504*	0.798	0.779	0.431	0.774
SC/ST	(1.115)	(0.300)	(0.711)	(0.815)	(0.967)	(0.562)
Received Loan	-2.018***	-0.099	-0.991*	-0.552	-0.624	0.735**
	(0.598)	(0.201)	(0.545)	(0.602)	(0.491)	(0.374)
SC/ST	-0.206	0.200*	0.051	0.196	-0.195	-0.251*
	(0.319)	(0.109)	(0.233)	(0.279)	(0.267)	(0.149)
Constant	0.519	-0.042	-1.710**	-1.149	5.406***	2.066***
	(0.929)	(0.308)	(0.833)	(1.115)	(0.812)	(0.390)
Sharpened q-values	0.493	0.493	0.493	0.493	0.511	0.493
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan *	-0.985	-0.579	-0.282	0.495	1.726*	0.994
SC/ST	(1.335)	(0.724)	(0.832)	(1.057)	(0.968)	(0.627)
Received Loan	0.605	-0.258	-0.199	-0.447	-0.658	-0.227
	(0.693)	(0.192)	(0.501)	(0.640)	(0.482)	(0.307)
SC/ST	-0.239	0.238	-0.059	-0.154	-0.293	-0.272
	(0.420)	(0.307)	(0.266)	(0.298)	(0.209)	(0.199)
Constant	2.321***	-0.141	1.277	4.390***	2.852***	2.401***
	(0.869)	(0.393)	(0.919)	(1.390)	(0.729)	(0.526)
Sharpened q-values	0.852	0.852	0.961	0.961	0.513	0.513
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The exogenous variable treatment assignment is interacted with binary variable of household belonging to Scheduled Caste/ Scheduled Tribe to instrument for heterogeneous effect of caste in loan receiving households. Dependent variables are number of hours spent in activity and listed as column titles. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. ∗ p< 0.1, ∗∗ p< .05, ∗ ∗ ∗ p< .01

Table A.38: LATE: Heterogeneous effects of caste on participation in decisions

	Share of decisions made (1)	Farming (2)	Livestock & Poultry (3)	Non agriculture (4)	Household purchases (5)	Private consumption (6)	Domestic work (7)	Personal activities (8)
Panel A: Women								
Received loan *	0.124	0.311*	-0.064	0.140	0.269	0.230	0.208	0.092
SC/ST	(0.109)	(0.183)	(0.156)	(0.173)	(0.194)	(0.190)	(0.202)	(0.171)
Received Loan	0.007	-0.315**	0.087	-0.190	0.082	0.079	0.129	-0.132
	(0.080)	(0.135)	(0.091)	(0.132)	(0.134)	(0.124)	(0.142)	(0.121)
SC/ST	0.039	0.005	0.065*	0.051	0.077	0.084	0.046	0.022
	(0.032)	(0.069)	(0.039)	(0.052)	(0.061)	(0.068)	(0.062)	(0.057)
Constant	-0.049	0.150	0.002	-0.348	-0.057	0.073	0.120	1.016***
	(0.114)	(0.204)	(0.122)	(0.229)	(0.200)	(0.199)	(0.215)	(0.160)
Sharpened q-values	0.941	0.941	0.941	0.941	0.941	0.941	0.941	0.941
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan *	-0.022	0.019	0.054	-0.037	0.028	-0.188	-0.223	-0.045
SC/ST	(0.101)	(0.175)	(0.164)	(0.184)	(0.204)	(0.230)	(0.215)	(0.174)
Received Loan	0.028	-0.059	-0.130	-0.022	-0.002	0.052	0.150	0.150*
	(0.054)	(0.081)	(0.096)	(0.110)	(0.128)	(0.124)	(0.133)	(0.085)
SC/ST	0.007	-0.042	-0.016	-0.023	-0.047	0.113	0.069	0.008
	(0.026)	(0.068)	(0.056)	(0.053)	(0.060)	(0.073)	(0.070)	(0.056)
Constant	0.147*	0.625***	-0.051	0.227	0.757***	0.291*	0.669***	0.160
	(0.076)	(0.155)	(0.130)	(0.207)	(0.195)	(0.163)	(0.145)	(0.171)
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables are listed as column titles. The exogenous variable treatment assignment is interacted with binary variable of household belonging to Scheduled Caste/ Scheduled Tribe to instrument for heterogeneous effect of caste in loan receiving households. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.39: Baseline characteristics by loan application status

	(1) All	(2) Treatment - Received loan	(3) Treatment - Denied loan	(4) Treatment - Didn't apply	(5) Control	F-test for balance across all groups	(2)-(5) Mean Difference	(3)-(5) Mean Difference	(4)-(5) Mean Difference
Respondent's age	43.301	43.168	42.523	44.929	42.727	3.108**	0.441	-0.204	2.202**
	(-0.447)	(-0.727)	(-1.817)	(-0.725)	(-0.469)	0.031			
Respondent is female	0.367	0.349	0.326	0.406	0.363	0.514	-0.013	-0.037	0.044
	(-0.042)	(-0.061)	(-0.112)	(-0.052)	(-0.067)	0.674			
Respondent is married	0.934	0.94	0.907	0.952	0.927	1.042	0.014	-0.02	0.025
	(-0.01)	(-0.015)	(-0.034)	(-0.011)	(-0.018)	0.379			
Can read and write	0.875	0.872	0.988	0.839	0.88	13.875***	-0.007	0.109***	-0.041
	(-0.016)	(-0.021)	(-0.011)	(-0.033)	(-0.023)	0			
Completed at least primary school	0.788	0.79	0.895	0.71	0.809	4.791***	-0.019	0.086*	-0.099**
	(-0.022)	(-0.036)	(-0.034)	(-0.039)	(-0.031)	0.004			
Completed at least	0.457	0.432	0.558	0.352	0.505	2.841**	-0.073	0.053	-0.154**
secondary school	(-0.029)	(-0.042)	(-0.089)	(-0.049)	(-0.044)	0.043			
Respondent belongs	0.358	0.335	0.198	0.506	0.323	7.771***	0.012	-0.125	0.183***
to SC/ST category	(-0.035)	(-0.051)	(-0.085)	(-0.051)	(-0.046)	0			
Annual Household	0.474	0.494	0.291	0.59	0.433	3.333**	0.061	-0.142	0.157*
Income(<90,000)	(-0.043)	(-0.067)	(-0.098)	(-0.061)	(-0.063)	0.024			
Annual Household Income	0.237	0.259	0.244	0.206	0.239	0.327	0.019	0.005	-0.033
(90,000 - 120,000)	(-0.023)	(-0.044)	(-0.067)	(-0.038)	(-0.034)	0.806			
Annual Household	0.289	0.247	0.465	0.203	0.327	3.982**	-0.08	0.138	-0.124*
Income(>120,000)	(-0.038)	(-0.058)	(-0.085)	(-0.049)	(-0.053)	0.011			
Owns land	0.479	0.483	0.314	0.555	0.464	2.637*	0.019	-0.15	0.091
	(-0.037)	(-0.065)	(-0.107)	(-0.039)	(-0.054)	0.055			
Asset quintile	2.986	3.108	2.488	3.123	2.924	1.796	0.184	-0.435	0.199
	(-0.114)	(-0.201)	(-0.352)	(-0.13)	(-0.169)	0.155			
Respondent has bank	0.933	0.946	0.988	0.894	0.937	7.711***	0.009	0.052***	-0.043*
account	(-0.01)	(-0.016)	(-0.012)	(-0.019)	(-0.014)	0			
Borrowed money last	0.439	0.474	0.512	0.329	0.463	2.451*	0.012	0.049	-0.134**
12 months	(-0.034)	(-0.062)	(-0.103)	(-0.044)	(-0.05)	0.07			
Has outstanding loan	0.212	0.23	0.244	0.213	0.198	0.089	0.032	0.046	0.015
	(-0.035)	(-0.06)	(-0.102)	(-0.053)	(-0.048)	0.966			
F-test of joint significance (F-stat)							0.683	1.044	2.116**
N	1429	352	86	310	681				

Notes: The table reports averages of respondent and household characteristics from the baseline survey by category of application status. Column 1 reports pooled estimates for the full sample. Column 2 reports the averages for households in the loan treatment arm that applied for the loan and received it. Column 3 describes households in treatment that were unsuccessful in getting the loan and column 4 includes households in treatment that did not apply for the loan. Column 5 describes households in the control arm. The fifth column reports the f test for whether each variable is balanced across arms and the last column provides coefficients of a t-test comparing the treatment and control averages. "SC/ST" is abbreviation for Scheduled Caste/ Scheduled Tribe. Standard errors are clustered at the level of treatment assignment (village) and reported below mean values for each variable. The number of observations are included in the last row. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.40: LATE: Independent bank account and debit card use re-weighted by propensity to borrow

	Own bank account	Use debit card alone	Share debit card
	(1)	(2)	(3)
	Panel A: Women		
Received loan product	-0.139	0.740***	-0.154
	(0.141)	(0.116)	(0.117)
Constant	0.570**	0.907***	0.022
	(0.273)	(0.163)	(0.256)
Sharpened q-values	0.273	0.001	0.237
Observations	1126	968	968
Control group mean	0.857	0.183	0.318
	Panel B: Men		
Received loan product	0.070	0.129	0.245**
	(0.095)	(0.112)	(0.118)
Constant	0.618***	1.715***	-0.351**
	(0.167)	(0.205)	(0.172)
Sharpened q-values	0.443	0.334	0.125
Observations	1364	1207	1207
Control group mean	0.836	0.524	0.236
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on bank account ownership and use of debit card. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. Columns 2 and 3 are conditional on respondent owning an individual bank account. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.41: LATE: Mobile phone and digital financial services re-weighted by propensity to borrow

	Regular mobile	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)	(4)
Panel A: Women				
Received loan product	0.177	0.227*	0.936***	0.917***
	(0.143)	(0.124)	(0.115)	(0.103)
Sharpened q-values	0.099	0.047	0.001	0.001
Observations	625	663	968	968
Control group mean	0.322	0.287	0.213	0.173
Panel B: Men				
Received loan product	-0.070	-0.046	0.512***	0.475***
	(0.116)	(0.110)	(0.088)	(0.086)
Constant	-0.059	0.911***	1.596***	1.631***
	(0.176)	(0.211)	(0.179)	(0.178)
Sharpened q-values	0.508	0.508	0.001	0.001
Observations	732	874	1207	1207
Control group mean	0.629	0.730	0.610	0.579
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on mobile phone ownership and uptake of digital financial services. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles. The analysis is conditional on respondent owning an individual bank account. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.42: LATE: Time use (peak agricultural season) re-weighted by propensity to borrow

	Farming (1)	Livestock & Poultry (2)	Non-agriculture (3)	Household purchases (4)	Domestic work (5)	Personal activity (6)
Panel A: Women						
Received loan product	-2.051**	-0.454*	-2.222***	-0.160	-0.590	1.156***
	(0.912)	(0.263)	(0.649)	(0.235)	(0.683)	(0.339)
Constant	2.307*	0.568	-1.687	-1.139**	5.140***	1.090*
	(1.225)	(0.493)	(1.217)	(0.525)	(0.957)	(0.593)
Sharpened q-values	0.035	0.068	0.004	0.198	0.184	0.004
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan product	0.445	-0.855*	-1.247*	-0.652***	1.507*	0.578**
	(0.984)	(0.499)	(0.726)	(0.221)	(0.846)	(0.228)
Constant	3.527***	-0.539	0.725	0.596	1.091	2.076***
	(1.173)	(0.536)	(1.208)	(0.485)	(0.945)	(0.478)
Sharpened q-values	0.122	0.075	0.075	0.019	0.075	0.029
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on time use in a busy day of the agricultural season. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles (see Table 2 for description of the dependent variables). Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.43: LATE: Time use (yesterday) re-weighted by propensity to borrow

	Farming	Livestock & Poultry	Non-agriculture	Household purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan product	-1.559**	-0.632*	-0.755	0.194	-0.431	1.251***
	(0.660)	(0.355)	(0.558)	(0.700)	(0.565)	(0.407)
Constant	-0.003	0.192	-1.875	-1.609	4.594***	1.900***
	(1.117)	(0.488)	(1.149)	(1.164)	(0.924)	(0.597)
Sharpened q-values	0.048	0.112	0.177	0.359	0.359	0.013
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan product	-0.263	-0.965*	-0.747	-0.627	-0.269	0.205
	(0.812)	(0.557)	(0.673)	(0.751)	(0.619)	(0.342)
Constant	2.262**	-0.477	-0.128	3.441**	2.685***	3.543***
	(1.003)	(0.614)	(1.128)	(1.574)	(0.750)	(0.649)
Sharpened q-values	1	0.993	1	1	1	1
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on time use on day prior to survey (yesterday). Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles (see Table 2 for description of the dependent variables). Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table A.44: LATE: Involvement in decision-making re-weighted by propensity to borrow

	Share of decisions made (1)	Farming (2)	Livestock & Poultry (3)	Non agriculture (4)	Household purchases (5)	Private consumption (6)	Domestic work (7)	Personal activities (8)
Panel A: Women								
Received loan product	0.066	-0.245*	0.041	-0.164	0.213	0.141	0.267**	-0.021
	(0.075)	(0.137)	(0.108)	(0.139)	(0.139)	(0.115)	(0.123)	(0.123)
Constant	-0.063	0.263	0.094	-0.420*	-0.042	0.050	0.158	0.954***
	(0.121)	(0.211)	(0.150)	(0.243)	(0.223)	(0.205)	(0.236)	(0.182)
Sharpened q-values	0.513	0.344	0.62	0.413	0.413	0.413	0.303	0.62
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan product	-0.028	0.008	-0.140	-0.143	-0.132	-0.134	-0.053	0.047
	(0.069)	(0.129)	(0.128)	(0.144)	(0.137)	(0.131)	(0.121)	(0.145)
Constant	0.008	0.734***	-0.149	-0.020	0.638***	-0.014	0.525***	0.018
	(0.096)	(0.172)	(0.142)	(0.243)	(0.204)	(0.161)	(0.169)	(0.191)
Sharpened q-values	1	1	1	1	1	1	1	1
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on involvement in production and non-production decisions. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for the household's receipt of loan product. The dependent variables (binary) are listed as column titles (see Table 2 for description of the dependent variables). Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01